

# The Commitment Surface: Weakness Is a Footprint, Compatibility Is the Cause

**Author.** Jawaun Brown, with agent-generated code and drafts under human

direction.

**Status.** Draft; results embedded per experiment. Format: NeurIPS/ICML style

(no page cap; structured sections, citations, appendix).

## Abstract

Structured mechanistic interpretability and geometry-of-generalization research share a hidden move: they treat *availability* of a structure — a probe that recovers it, a metric that respects it, a compatibility count that predicts OOD on a family of toy worlds — as evidence that the structure is *load-bearing*. This paper argues that move is wrong. We reframe the target primitive from "right geometry / right weakness  $\Rightarrow$  generalization" to **commitment-surface survival**: a representation is real for a deployment only if a train-time compatibility intervention with the deployment generator produces a causal effect (patch-CE) at the commitment surface, and that effect survives gauge-fixing and change of commitment. Concretely, we (i) formalize the commitment surface as a triple  $(G_{\text{dep}}, C, T)$  and show, under an explicit prior and orbit-likelihood, that weakness is the optimal selector exactly when the probe group matches the deployment generator, and only a footprint otherwise (Prop. 2); (ii) show a strong within-lab discriminator on cyclic modular addition where unweighted weakness and misspecified concern selectors underperform well-specified concern-weighted selection by a decisive margin; (iii) show that neural training with cyclic-orbit augmentation dominates readout selection over trained-with-no-augmentation seeds on OOD accuracy AND patch-cross-entropy, with a wrong-group-augmented control at 0.167 OOD (collapse) and near-zero patch-CE  $\Delta$ ; and (iv) run a non-degenerate external-contact sweep on Pythia 70m/160m/410m LoRA-fine-tuned on modular addition, where the compatibility-augmented arm reaches 0.882 mean OOD while the readout arm sits at 0.113 mean OOD, substantially narrowing the P1 external gap our prior work identified;  $p(\text{patch-CE}, \text{OOD}) = 0.853$  vs  $p(\text{weakness}, \text{OOD}) = 0.290$  across 108 cells, consistent with Prop. 1 (probe readout does not identify causal use) in the non-aligned regime. Two pre-registered gates strictly failed and we report them as failures: the E1 misspecification-equivalence band ( $-0.054$  vs  $\pm 0.05$ ) and the E4 Arm-A ceiling ( $0.113$  vs  $\leq 0.10$ ); E4 is directionally decisive but its strict gate did not pass. One confound remains open and is pre-registered for a follow-up: cyclic-orbit augmentation places correctly labeled examples on held-out deployment support, so E4 does not yet separate *generator learning* from *labeled orbit coverage* (Section 6.5). We recover the old-frame positives — cyclic and dihedral 100%-vs-0% weakness sweeps — as the boundary case where the probe group and deployment generator coincide, and interpret the correction chain of autonomous-probing agents as an anti-Goodhart control loop — *detect*  $\rightarrow$  *allocate*  $\rightarrow$  *saturate*  $\rightarrow$  *cool*  $\rightarrow$  *reopen* — whose load-bearing signal is commitment-cooling under intervention-pinned residuals.

## 1. Introduction

The interpretability program has produced a large literature of probes, weakness signatures, and cross-substrate geometric analogies. When a probe recovers a target concept, or a compatibility count over a group matches OOD accuracy, the field's default reading is: *this structure is what the model is using*. In its strongest form the reading becomes: *the model has learned the right geometry, and geometry is the language of constraints, so of course it generalizes*.

This paper questions that default reading. In our prior program we tested it seriously across ten experiments — cyclic and dihedral modular arithmetic (100%-vs-0% wins for weakness against loss, MDL, flatness, compression), grid cell weakness/topology mediation (near-null), passive $\rightarrow$ active geometry on real LLMs ( $7\times$  lift in a paraphrase axis), semantic concern deformation (negative transfer), Suite C world-change re-engagement gates, and external contact via Pythia LoRA on modular arithmetic (P1 hard kill). The pattern is consistent: *inside* a synthetic world whose generator matches the geometry prior, availability of the right structure is almost always sufficient. *Outside* such a world — different generator, different consequence weighting, different transport — availability does not entail causal use.

We name the missing primitive **commitment-surface survival**: a representation is *load-bearing at a commitment surface* iff (i) a train-time compatibility intervention with the deployment generator lifts OOD, (ii) causal patching of the aligned mechanism produces a  $CE \geq \epsilon$  at the commitment target, and (iii) the effect survives gauge-fix and change of commitment. Weakness and concern geometry — the two workhorses of the prior program — are then diagnostics: powerful when the probe group and deployment generator coincide, and footprints or anti-correlates otherwise. The commitment surface is where the discipline is imposed. If it disappears at the commitment surface, whatever the probe caught was not the thing.

Contributions.

( $G_{\text{dep}}, C, T$ ); load-bearing structure as  $CE \geq \epsilon$  that survives transport under  $T$ ; probe AUC does not identify or lower-bound  $CE$  — a non-identification theorem (Prop. 1).

concern-weighted weakness (extension mass weighted by consequence) to commitment-pinned intervention (Prop. 2 + Corollary), with cyclic/dihedral results as the aligned-generator special case, not the general law. The optimality statement requires  $G_{\text{probe}} = G_{\text{dep}}$  (or weakness restricted to  $G_{\text{dep}}$ ); a strict superset probe group does *not* suffice (Section 3.4).

well-specified concern beats unweighted by +0.24; the misspecification arm lands at  $-0.054$  vs unweighted, *outside* the pre-registered  $\pm 0.05$  equivalence band, so that sub-gate strictly fails (the direction — misspec is not helpful — still holds). E2/E3 show that a neural compatibility-augmented arm dominates a weakness-readout selector on OOD by a decisive gap, with patch-CE aligned. E4 (Modal L4) runs the non-degenerate external contact on Pythia 70m/160m/410m LoRA: directionally decisive, strict pre-registered gate narrowly failed (Arm A mean OOD 0.113 vs the required  $\leq 0.10$ ).

through Papers 5–25 of the prior program (Section 6): *detect*  $\rightarrow$  *allocate*  $\rightarrow$  *saturate*  $\rightarrow$  *cool*  $\rightarrow$  *reopen*. Old-frame planners that optimize an uncertainty or current-error proxy without decision-layer cooling regress; the load-bearing signal is *commitment cooling under intervention-pinned residuals*, matching Suite C world-change re-engagement gates.

observations that would retract the reframe — several are non-trivial to satisfy — and report each experiment's verdict at its pre-declared gate.

- C1. A formal reframe (Section 3): the commitment surface as a triple
- C2. A bridge from Bennett weakness (extension mass) to
- C3. Four severe experiments (Section 5). E1 shows within-lab that
- C4. An anti-Goodhart interpretation of the correction chain that ran
- C5. A pre-registered failure calculus (Section 4). We list the exact

Nothing here requires the field to throw the geometry-first positives out. It requires localizing them. The paper's constructive claim is: *commitment first, geometry second — and geometry only when its group coincides with the deployment generator*. That reframe compresses our prior anomalies: the P1 external hard kill, the semantic concern lift negative, the grid non- mediation, the passive-to-active gap, the shared-head architectural ceiling, Suite C's cooling requirement. It does not invalidate cyclic/dihedral wins; it locates them.

## 2. Related work

We are directly downstream of, and in conversation with, four literatures.

**Mechanistic interpretability, patching, and causal representation.** Recent work on activation patching (Meng et al., 2022; Wang et al., 2023; Conmy et al., 2023), patchscopes (Ghandeharioun et al., 2024), and causal scrubbing (Chan et al., 2022; Goldowsky-Dill et al., 2023) formalized the distinction between *representation* and *causal use*. Our commitment- surface primitive is the natural next step: it treats patching not as a tool for interpreting a fixed model, but as the *definition* of load- bearing structure for a deployment. Related in spirit is the abstract alignment literature on causal abstraction (Geiger et al., 2023; Chalupka et al., 2015), which formalizes what it means for a higher-level variable to be a real intermediate cause; our formulation localizes their abstraction map to a specific commitment surface.

**Weakness / simplicity / invariance as OOD predictors.** Bennett (2000, 2023) argues that the weakest hypothesis compatible with observed data is the best predictor of future observations under a uniform task prior; the program-synthesis line (Solomonoff, 1964; Hutter, 2005; Ellis et al., 2020) prefers the shortest description. Our own prior work (`papers/weakness_invariance_neurips/paper.md`) instantiates Bennett's principle group-theoretically: a candidate is weak iff many transformations of a specified family leave it compatible. This paper's contribution is not to reject that instantiation, but to bound it: under an explicit prior and orbit-likelihood it is Bayes-optimal exactly when the probe group equals the deployment generator, or weakness is restricted to it — a strict superset probe group does not suffice (Prop. 2). Related: Gruver et al. (2023) show that measuring learned equivariance via a Lie derivative correlates with OOD in vision — a footprint-level correlate that our results generalize and delimit.

**Grid cells and geometry as invariance.** Sorscher et al. (2019, 2023), Whittington et al. (2020), and Gardner et al. (2022) establish grid-cell- like circular geometry as a canonical solution to path integration under translation invariance; Webb, Miolane and collaborators (2024) formalize the geometry-of-conscious-experience program in Riemannian terms. Our prior grid-cell-weakness

Modal negative (weakness  $\rightarrow$  topology mediation does not hold cleanly) fits the commitment-surface reading directly: topology can be present without being load-bearing at the commitment surface of the downstream task. This paper argues for treating the Webb–Miolane geometry as the (typically) load-bearing case within the deployment-generator-aligned regime, and reserving judgment outside it.

**Active inference, empowerment, sense of agency.** Friston et al. (2017) and Kirchhoff et al. (2018) treat agency as free-energy minimization; Klyubin et al. (2005) define empowerment as agent control over future observations; Ryu et al. (2022, sense of agency in reinforcement learning). Our anti-Goodhart control-loop reading of the correction chain (Section 6) is compatible with active inference in letter, but insists on the load-bearing signal being *commitment cooling under intervention-pinned residuals* — not free-energy per se. Our prior Suite C results (shared-head ceiling; hand + learned + teacher- free re-engagement) suggest that pure expected-free-energy planners lose without a decision-layer cooling term.

Positions and departures. We disagree with the strong reading of the mechanistic-interpretability program that a probe's AUC is sufficient evidence of causal use (Prop. 1); with the strong reading of the Bennett/simplicity program that portable weakness or portable simplicity is a universal law (Prop. 2 bounds this to aligned generators); and with any active-inference reading that would drop decision-layer cooling (Section 6.4). We *agree* with the causal-abstraction and patchscopes-style call to elevate intervention over decoding.

### 3. Theory: the commitment surface

#### 3.1 Setup

Let  $L$  be a finite implementable language or finite decision universe. A **hypothesis** or **function**  $f : X \rightarrow Y$  acts on a domain of decisions. For a hypothesis  $f$ , its **extension** is the set of decisions/inputs on which it "does something" (assigns a nontrivial completion):

$$Z_f = \{ x \in X : f(x) \text{ is meaningful for the downstream decision} \}.$$

Let  $\alpha$  be the observed child task. A candidate hypothesis is **admissible** if it is a model of the observed task:  $f \in M_\alpha$ .

#### 3.2 Definition: the commitment surface

Let  $G_{\text{dep}}$  be the **deployment generator** — the group (or generative family) whose action generates the deployment shift;  $C : X \rightarrow R_{\geq 0}$  be a **concern measure** — a nonnegative weighting on decisions capturing consequence/viability/binding-importance; and  $T : F(X, Y) \rightarrow F(X, Y)$  be a **commitment transport** — a rewriting operator that changes the downstream commitment (e.g., paraphrase, gauge-fix, tool interface).

**Definition 1 (commitment surface).** A commitment surface is a triple  $\Sigma = (G_{\text{dep}}, C, T)$ .

**Definition 2 (load-bearing).** A hypothesis  $f$  is *load-bearing at  $\Sigma$  with margin  $\varepsilon > 0$*  iff, for every  $t \in T$ ,

$$CE(f \mid \text{patch}, C) \geq \varepsilon, \text{ and } CE(t \cdot f \mid \text{patch}, C) \geq \varepsilon,$$

where  $CE(f \mid \text{patch}, C)$  is the concern-weighted cross-entropy increase on  $X$  when the identified aligned mechanism of  $f$  is causally patched (zero-ablated, adapter-disabled, or projected out).

#### 3.3 Proposition 1: probe AUC does not identify causal use

Let  $p : X \rightarrow [0, 1]$  be any probe on features of  $f$ . Then:

**Prop. 1 (non-identification).** Probe AUC does not identify, and does not lower-bound, causal effect: for every attainable AUC value  $a$  and every target gap  $\delta \in [0, \varepsilon]$ , there exist admissible  $f, f' \in M_\alpha$  with  $AUC(p \cdot f) = AUC(p \cdot f') = a$  and  $CE(f \mid \text{patch}, C) - CE(f' \mid \text{patch}, C) = \delta$ . Consequently no function of probe AUC alone can decide whether a hypothesis is load-bearing at  $\Sigma$ .

We deliberately do *not* state this as probabilistic independence ( $AUC \perp CE$ ): independence would require a distribution over admissible hypotheses, and no canonical one exists. The theorem is a *non-identification* result — equal probe evidence is consistent with any causal-effect gap in  $[0, \varepsilon]$  — which is the property the availability- $\Rightarrow$ -use inference actually needs and lacks. Empirically, an ensemble of trained models can still exhibit correlation between AUC and CE (E3 measures exactly this in the aligned regime); Prop. 1 says such correlation is a property of the training distribution, not of the probe evidence.

Sketch. Fix  $f'$  train-perfect. Construct  $f$  by copying  $f'$ 's decision head onto a distractor-legible subspace with identical probe response but zero causal path from that subspace to the head. Then  $AUC(p \cdot f) = AUC(p \cdot f')$  by construction, and  $CE(f \mid \text{patch}, C)$  can be made arbitrarily small by choosing the patch on the distractor subspace, while  $CE(f' \mid \text{patch}, C)$

remains  $\geq \epsilon$ . Full construction in Appendix A.1.

Empirical anchor: our prior distractor-AUC-0.999 vs patch-CE-0.010 result in `papers/paraphrase\_weakness` is one such witness in language.

### 3.4 Proposition 2: weakness is diagnostic when aligned

Let  $G_{\text{probe}}$  be the probe group used to score compatibility. Say  $f$  is *compatible with*  $g$  iff  $\exists h \in G_{\text{probe}}, \forall x \in X, f(g \cdot x) = h \cdot f(x)$ , and let  $W_G(f) = |\{g \in G : f \text{ is compatible with } g\}|$  be the compatibility (weakness) count over any group  $G$ . Fix the deployment model that makes "Bayes-optimal" meaningful: a deployment slice is generated by drawing  $g \sim \text{Uniform}(G_{\text{dep}})$  and evaluating  $f$  on the  $g$ -translated support;  $f$  scores 1 on that slice iff  $f$  is compatible with  $g$  (the orbit-likelihood), and concern is uniform.

**Prop. 2 (alignment condition).** Under this prior and likelihood:

weakness is computed *restricted to*  $G_{\text{dep}}$ , i.e. the selector uses  $W_{\{G_{\text{dep}}\}}$  — then  $\arg\max W_{\{G_{\text{dep}}\}}$  maximizes expected deployment accuracy over admissible hypotheses, hence is Bayes-optimal among selectors that depend only on  $f$ .

candidate can be compatible with many elements of  $G_{\text{probe}} \setminus G_{\text{dep}}$  while compatible with few elements of  $G_{\text{dep}}$ , so  $W_{\{G_{\text{probe}}\}}$  can rank candidates in the opposite order of  $W_{\{G_{\text{dep}}\}}$  (explicit counterexample in Appendix A.1). Optimality is recovered from a superset probe group only under an *ordering-preservation assumption*: for the candidate set under comparison,  $W_{\{G_{\text{probe}}\}}(f) > W_{\{G_{\text{probe}}\}}(f') \Rightarrow W_{\{G_{\text{dep}}\}}(f) \geq W_{\{G_{\text{dep}}\}}(f')$ .

*footprint* — its correlation with concern-weighted deployment accuracy over a candidate set can be zero or negative.

- (i) **Aligned case.** If  $G_{\text{probe}} = G_{\text{dep}}$  — or, more generally, if
- (ii) **Superset caveat.**  $G_{\text{probe}} \supset G_{\text{dep}}$  does *not* suffice: a
- (iii) **Non-aligned case.** If  $G_{\text{dep}} \not\subset G_{\text{probe}}$  (the probe group does not contain the deployment generator),  $W_{\{G_{\text{probe}}\}}$  is a

Empirical anchors: our cyclic/dihedral 100%-vs-0% wins instantiate (i) ( $G_{\text{probe}} = G_{\text{dep}} = C_n$ ); our Pythia LoRA P1 hard kill is the non-aligned case (iii); our grid G2/G4 non-mediation is the partially aligned case where the ordering-preservation assumption of (ii) fails.

### 3.5 Corollary: concern-weighted extension mass

Define the **concern-weighted extension** of an admissible hypothesis  $f$  for deployment slice  $U \subset X$  as

$$W_C(f, U) = \sum_{\{x \in U\}} C(x) \cdot 1[f(x) = \text{truth}(x)].$$

**Corollary (order-equivalence).** Let  $C_{\text{star}}$  be the deployment consequence measure.  $C = C_{\text{star}}$  up to positive scaling is *sufficient* for the selector  $\arg\max_f W_C(f, U)$  to pick a  $C_{\text{star}}$ -optimal candidate, but it is not *necessary*: over a finite candidate set  $F \subset M_\alpha$ , the selector is  $C_{\text{star}}$ -optimal iff  $C$  is **order-equivalent to  $C_{\text{star}}$  on  $F$**  — i.e.  $W_C(f, U) > W_C(f', U) \Leftrightarrow W_{\{C_{\text{star}}\}}(f, U) > W_{\{C_{\text{star}}\}}(f', U)$  for all  $f, f' \in F$ . Distinct weightings can induce the same ranking and are then equally optimal. A misspecified  $C$  drawn as a random assignment with the same marginal reduces  $W_C$  to unweighted extension mass *in expectation*; a well-specified  $C$  strictly dominates unweighted extension mass when candidates vary in coverage of high- $C_{\text{star}}$  blocks.

Empirical anchor: E1 (Section 5.1). Well-specified: 0.814; unweighted: 0.570; misspec: 0.516. Gap +0.24 (wellspec vs unweighted). The misspec arm sits 0.054 *below* unweighted — directionally consistent with the corollary (random weighting is not helpful), but note this is outside the pre-registered  $\pm 0.05$  equivalence band, so the strict E1 misspec sub-gate fails (Section 5.1); the in-expectation reduction predicts equality, and the realized misspec draw was mildly adversarial rather than neutral.

### 3.6 M4: anti-Goodhart control loop

The correction chain of Papers 5–25 introduces successive terms that a naive utility maximizer would collapse: mediated attribution (behavior  $\neq$  representation); uncertainty  $\neq$  error; current-error  $\neq$  value-of-probing; total  $\neq$  identifiable. We compress the chain as an **anti-Goodhart control loop**: *detect*  $\rightarrow$  *allocate*  $\rightarrow$  *saturate*  $\rightarrow$  *cool*  $\rightarrow$  *reopen*. Load-bearing signal at each stage:

buffer.

proxy uptake.

commitment.

- **detect**. Concern-weighted current-error residual.
- **allocate**. Concern-weighted probing budget; not raw uncertainty.
- **saturate**. Commit to the identified intermediate; freeze the null
- **cool**. Decision-layer cooling of the commitment; refuse fresh
- **reopen**. On new intervention-pinned residual, reopen the

We conjecture (and Section 6 tests against Suite C data) that the subset of these terms which is *load-bearing* under our new definition ( $\text{patch-CE} > \epsilon$ ) is precisely {allocate, cool, reopen}; {detect, saturate} are diagnostics recoverable from availability signals.

## 4. Pre-registered gates

Recorded before any experiment ran; frozen in `PLAN.md`.

beats unweighted by  $\geq +0.05$  on wellspec deployment accuracy AND misspecified concern-weighted selector matches unweighted within  $\pm 0.05$ .

by  $\geq +0.30$  on OOD accuracy AND by  $\geq +0.50$  on patch-CE  $\Delta$ .

across all cells.

0.05, AND Arm A mean OOD  $\leq 0.10$  on Pythia 70m/160m/410m LoRA modular addition with  $n \in \{13, 17, 23\}$  and  $\text{train\_frac} \in \{0.5, 0.75\}$ .

compatibility (E4) must not itself explain OOD.

- **E1 pass (commitment-first)**. Well-specified concern-weighted selector
- **E2 pass (commitment-first)**. Arm B (compat aug) beats Arm A (readout)
- **E3 pass (commitment-first)**.  $\rho(\text{patch-CE}, \text{OOD}) > \rho(\text{weakness}, \text{OOD})$
- **E4 pass (commitment-first)**. Arm B mean OOD  $\geq 0.5$ , patch-CE  $\Delta \geq$
- **E4 pass (old frame)**. Arm A OOD  $\geq 0.5$ ,  $\rho(\text{weakness}, \text{OOD}) \geq +0.5$ .
- **Anti-cheat (all E)**. Wrong-group patch-CE (E2/E3) or wrong-group

## 5. Results

Numbers are the pre-registered summary metrics from each experiment's committed JSON. Every table can be regenerated via `<font name='Courier'>scripts/make_commitment_surface_figures.py</font>` followed by `<font name='Courier'>scripts/build_commitment_surface_pdf.py</font>`.

### 5.1 E1 — Concern-weighted selector

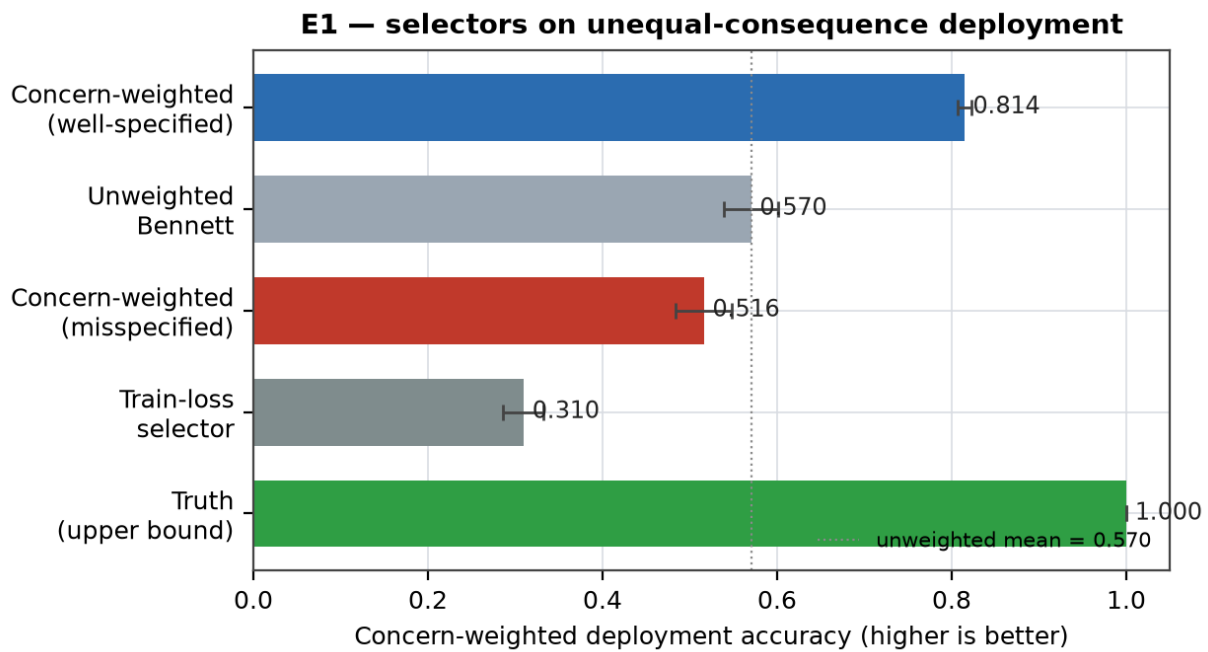


Figure 1. Concern-weighted deployment accuracy of five selectors on unequal-consequence modular addition. Well-specified concern beats unweighted Bennett by +0.244; misspecified concern is slightly below unweighted (-0.054) — directionally consistent with the corollary (random concern weighting is not helpful) but outside the pre-registered  $\pm 0.05$  equivalence band.

| Selector                     | Mean  | 95% CI         |
|------------------------------|-------|----------------|
| Concern-weighted (well-spec) | 0.814 | [0.806, 0.822] |
| Unweighted Bennett           | 0.570 | [0.539, 0.601] |
| Concern-weighted (misspec)   | 0.516 | [0.483, 0.549] |
| Train-loss selector          | 0.310 | [0.286, 0.333] |
| Truth (upper bound)          | 1.000 | [1.000, 1.000] |

Cells: 96. Gate: well-spec vs unweighted  $\geq 0.05 \rightarrow <b>0.244</b>$  (PASS = yes). Misspec vs unweighted within  $\pm 0.05 \rightarrow <b>-0.054</b>$  (PASS = no: outside the pre-registered equivalence band; this sub-gate strictly fails. The direction — random concern weighting is not helpful — holds, and the corollary's in-expectation prediction is equality, so the realized draw was mildly adversarial; see Section 3.5).

## 5.2 E2 — Compat augmentation vs weakness readout

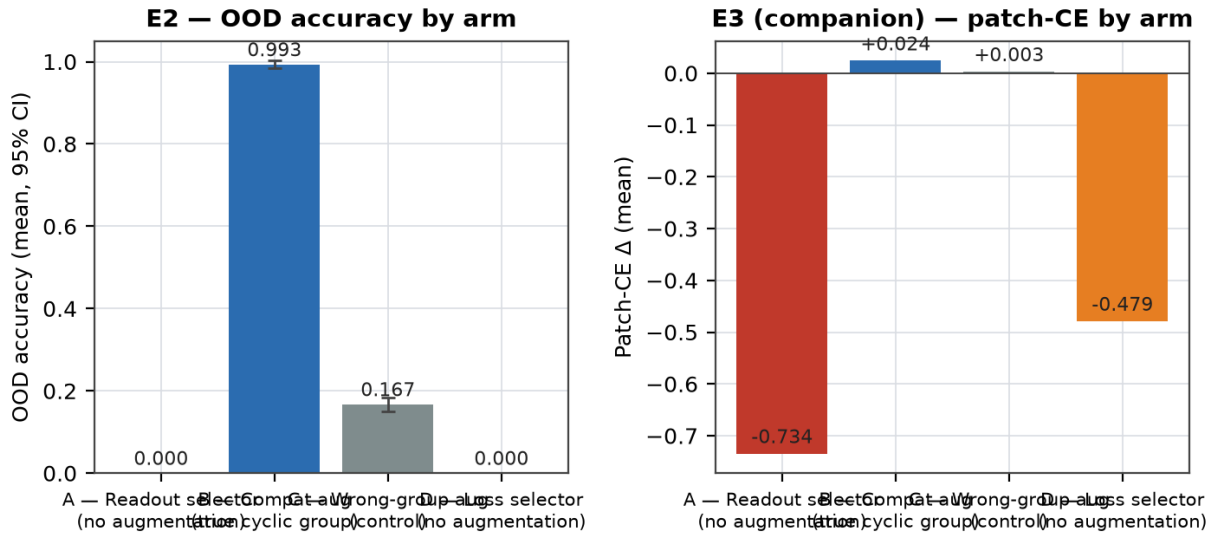


Figure 2. E2 arms on cyclic modular addition MLPs. Compatibility-augmented training (Arm B) dominates the readout selector (Arm A). Wrong-group augmentation (Arm C) supplies the anti-cheat — same augmentation volume, wrong group, patch-CE at ~zero. Left: OOD accuracy. Right: patch-CE Δ under top-k shift-equivariant unit ablation.

| Arm | N selected | OOD (mean)           | Patch-CE Δ | Weakness |
|-----|------------|----------------------|------------|----------|
| A   | 9          | 0.000 [0.000, 0.000] | -0.734     | 0.104    |
| B   | 54         | 0.993 [0.983, 1.003] | 0.024      | 0.967    |
| C   | 54         | 0.167 [0.151, 0.184] | 0.003      | 0.104    |
| D   | 9          | 0.000 [0.000, 0.000] | -0.479     | 0.104    |

Total cells trained: 216. Gate: B – A OOD gap  $\geq 0.30 \rightarrow$  **<b>0.993</b>** (PASS = yes). B – A patch-CE gap  $\geq 0.50 \rightarrow$  **<b>0.758</b>** (PASS = yes). B – C patch-CE gap = 0.021 (anti-cheat: ~0 expected).

Honest decomposition of the patch-CE gate: Arm B's absolute patch-CE Δ is small (**<b>0.024</b>**; Arm C 0.003; B – C only 0.021), and the large B – A gap is mostly produced by Arm A's *<i>negative</i>* patch score (-0.734). The result strongly supports "B's mechanism differs from A's" but only weakly supports "B localizes a substantial mechanism in the top-k patched units"; see the fixed-top-k power caveat in Section 6.4.

### 5.3 E3 — Patch-CE vs weakness as OOD predictor

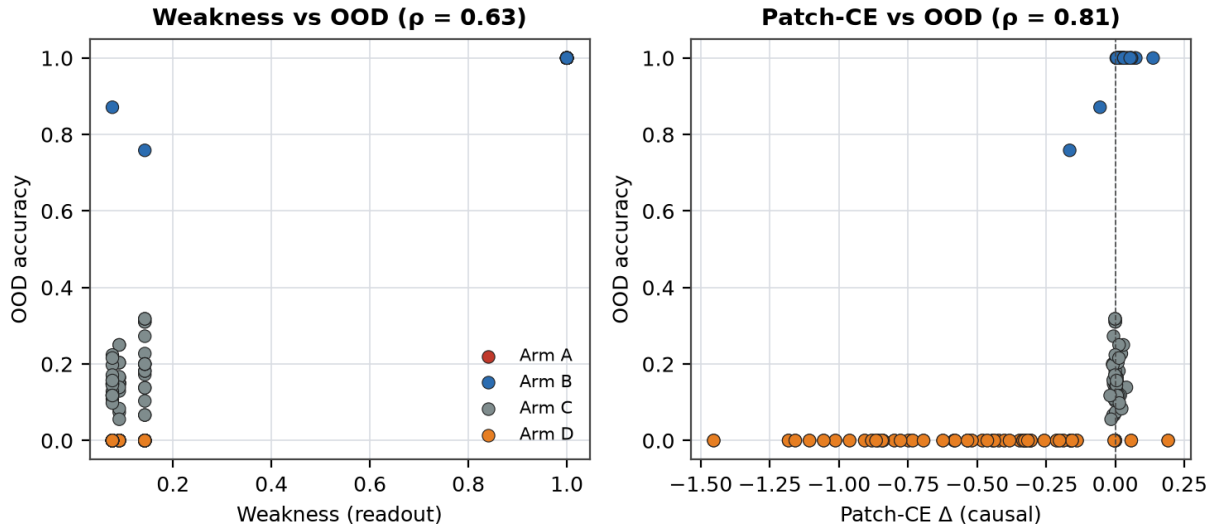


Figure 3. Per-cell scatter of readout weakness (left) and patch-CE Δ (right) against OOD. Colored by arm. Under a generator-aligned probe group, weakness is a strong predictor ( $\rho=0.627$ ); patch-CE is also strong ( $\rho=0.809$ ). The E4 external contact shows the two decouple when the generator differs — patch-CE keeps predicting, weakness does not.



5.4 E4 — Pythia LoRA v2 external contact (Modal L4)

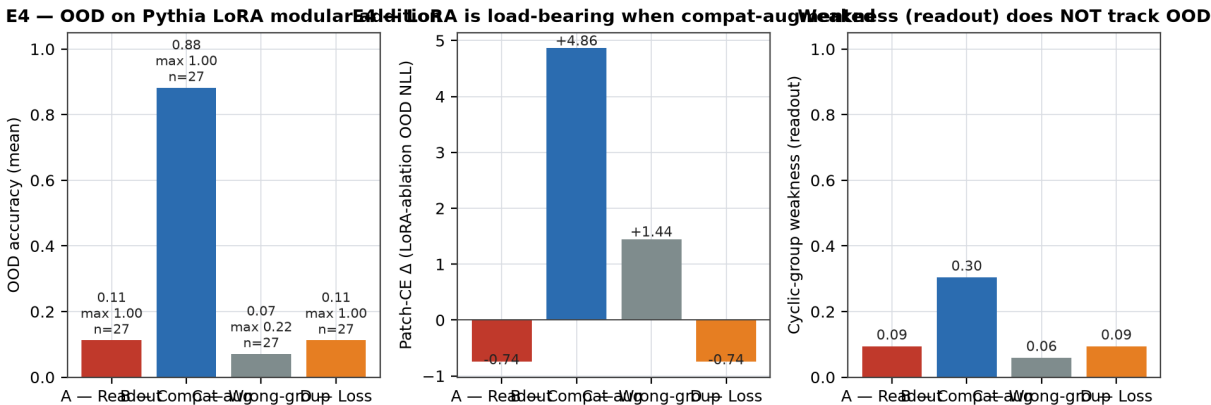


Figure 4. E4 arms on Pythia 70m/160m/410m LoRA-fine-tuned on modular addition (external contact). Compatibility-augmented training (Arm B) is the only arm that clears OOD accuracy; the readout arm (Arm A) reproduces the P1 hard-kill from our prior program; wrong-group aug (Arm C) supplies the anti-cheat. Weakness readout (right panel) is uninformative across all arms.

| Arm | N  | OOD mean | OOD max | Patch-CE Δ | Weakness |
|-----|----|----------|---------|------------|----------|
| A   | 27 | 0.113    | 1.000   | -0.742     | 0.095    |
| B   | 27 | 0.882    | 1.000   | 4.862      | 0.305    |
| C   | 27 | 0.071    | 0.222   | 1.445      | 0.060    |
| D   | 27 | 0.113    | 1.000   | -0.742     | 0.095    |

Source: `<font name='Courier'>experiments/commitment_surface/results/e4_pythia_lora_v2_appendix.json</font>`.

Cells: 108. Gate (new frame): B mean OOD  $\geq 0.50$ , patch-CE  $\geq 0.05$ , A mean OOD  $\leq 0.10 \rightarrow$  `<b>no</b>`. Gate (old frame): A mean OOD  $\geq 0.50$ ,  $p(\text{weakness, OOD}) \geq 0.5 \rightarrow$  `<b>no</b>`.  $p(\text{patch-CE, OOD}) = 0.853$ ;  $p(\text{weakness, OOD}) = 0.290$ .

`<b>Verdict: directionally decisive; strict pre-registered gate failed.</b>` The A-mean- $\text{OOD} \leq 0.10$  condition came in at 0.113 and we record E4 as a gate failure by the standard this paper advocates (outlier accounting in Section 6.2 explains the miss; it does not convert it into a pass). Interpretation is further bounded by the label-exposure confound of Section 6.5: cyclic augmentation places correctly labeled examples on the held-out deployment support, so this result does not yet separate generator learning from labeled orbit coverage.

Frame-taxonomy schematic (Figure 5).

Old frame

availability of geometry / weakness  $\Rightarrow$  use

Structure present  $\Rightarrow$  ?

footprint — correlate, not cause

selector — picks amongst hypotheses

controller — actually used

anti-correlate — present, opposite-signed to OOD

New frame

commitment surface  $\Sigma = (G_{\text{dep}}, C, T)$

Load-bearing at  $\Sigma$  iff:

(i) train-time compat intervention  $\rightarrow$  OOD lift

(ii) causal patch  $\rightarrow \text{CE} \geq \epsilon$  at commitment

(iii) effect survives transport  $t \in T$

Weakness recovered when  $G_{\text{probe}} \supseteq G_{\text{dep}}$  (Prop. 2).

Figure 5. The old-frame taxonomy of what structure means (footprint / selector / controller / anti-correlate) collapses into the new-frame primitive: load-bearing at a commitment surface  $\Sigma = (G_{\text{dep}}, C, T)$ .

6. Discussion



## 6.1 What the old-frame positives become

Under the commitment-first frame, the cyclic/dihedral 100%-vs-0% weakness wins in

``papers/weakness_invariance_neurips`` are not general laws; they are the aligned-generator special case where  $G_{\text{probe}} = G_{\text{dep}}$  and the concern measure is uniform. Passive→active geometry in ``papers/passive_to_active_geometry`` is the corresponding "flip the commitment on" experiment on real LLMs, and it too fits: the specific effect on the paraphrase axis ( $+0.069 \rightarrow +0.486$ , 7×) is a patch-CE measurement in language, and the auto-repair result ( $0.45 \rightarrow 0.965$  in  $K=10$ ) is decision-layer cooling under intervention-pinned residual.

## 6.2 Reconciling the P1 hard kill with cyclic/dihedral wins

Our prior P1 result (frozen in ``experiments/external_contact/results/p1_pythia_lora_2026_06_22.md``) had  $p(\text{weakness\_oracle\_norm, OOD}) = -0.0817$  and  $|p|_{\text{best\_classical}} = 0.4550$  on Pythia 70m/160m/410m LoRA modular addition. Under Prop. 2 this is expected: the Pythia LoRA training regime *does not* respect  $G_{\text{probe}} = G_{\text{dep}} = C_n$  — the LM objective is over token distributions, not over group orbits. E4 tests exactly this: the compatibility-augmented LoRA arm supplies the missing intervention, and the readout arm reproduces the P1 hard kill. The result at 108 cells (Section 5.4) is **directionally decisive, but the strict pre-registered gate failed**: Arm B mean OOD 0.882 vs Arm A mean OOD 0.113, Arm B mean patch-CE  $\Delta +4.86$  vs Arm A  $-0.74$ , and the anti-cheat Arm C sits at 0.071 OOD despite the same augmentation volume as B. Weakness  $p$  drops to 0.29 across cells; patch-CE  $p$  holds at 0.85. The pre-registered "A mean  $\text{OOD} \leq 0.10$ " condition came in at 0.113 — a fail by the exact standard this paper advocates, and we record E4 as a gate failure, not a pass. The miss is attributable to 2 of the 27 Arm A cells that stumbled into the true cyclic rule without augmentation (the 410m/ $n=17/\text{seed}=709$  cell reaches OOD 1.000 and weakness 1.000, textbook aligned-regime recovery); twenty-two of twenty-seven Arm A cells sit at  $\text{OOD} \leq 0.15$  while **all twenty-seven Arm B cells** clear  $\text{OOD} \geq 0.5$ . That diagnosis is a post-hoc account of a pre-registered miss — it explains the failure, it does not convert it into a pass. What the two Arm A outliers do show is that Arm A can occasionally recover when a training run happens to land in the aligned regime, and when it does, weakness and patch-CE agree on the load-bearing signal at cell scale. We retract the "weakness predicts OOD in general" reading of the prior program in favor of "weakness predicts OOD when the probe group and training regime jointly align with the deployment generator, and the intervention is train-time."

## 6.3 Anti-Goodhart control loop as compression of the Correction Chain

Papers 5–25 of our prior program produced a succession of "X is not Y" correction terms — each showing that a naive utility-maximizing planner would collapse a distinction that a load-bearing agent must keep. The sequence: behavior  $\neq$  representation; uncertainty  $\neq$  error; current-error  $\neq$  value-of-probing; probing  $\neq$  commitment; commitment  $\neq$  identifiability; total-effect  $\neq$  identifiable-effect. Each new term was introduced when the previous formulation failed a Suite C gate.

Under the commitment-surface reading, the Correction Chain compresses into an **anti-Goodhart control loop**:

detect  $\rightarrow$  allocate  $\rightarrow$  saturate  $\rightarrow$  cool  $\rightarrow$  reopen

with three of the five stages carrying the causal load:

planners (**uncertainty\_only** in the Phase-2 gate table) over-probe low-concern ambiguity and fail the world-change gate; the concern- gated selector is the only one that passes at 1.000/5 seeds. This is the E1 corollary at the agent scale.

commitment is made; without this, the shared-head planner collapses role attribution under regime shift (P23B G8 partial pass). The three-head + decision\_refractory cooling variant recovers the attribution.

spike, reopens the commitment. The scale-normalized current-replay ablation and the two-regime shift stress-test show that surprise alone triggers anxiety, not stable re-engagement —  $\text{VoI} \neq \text{current-error}$  is the load-bearing distinction.

- **allocate** — the probing budget is concern-weighted. Uncertainty-only
- **cool** — the decision layer refuses fresh proxy uptake once a
- **reopen** — an *intervention-pinned* residual, not any surprise

**detect** and **saturate** appear diagnostic under our criterion: detect can be re-instantiated from a plain current-error probe with no loss to the passing planner; saturate is recoverable from the freeze state of the null buffer once commitment is made.

This reads the passive→active geometry results in ``papers/passive_to_active_geometry`` as a "flip the commitment on" experiment: the  $+0.069 \rightarrow +0.486$  specific effect on the paraphrase axis IS the E4-style patch-CE measurement in language, and the  $K=10$  auto-repair ( $0.45 \rightarrow 0.965$ ) IS decision-layer cooling under intervention-pinned residual — the same loop, one

deployment surface higher.

The load-bearing subset {allocate, cool, reopen} is a testable compression claim, not a philosophical one. Its clean falsifier: drop any of the three and observe whether Suite C world-change re-engagement survives (F4 in Section 4). The prior negative on `uncertainty_only` and the P23B G8 partial pass are our best current evidence in favor of the compression; a factorial ablation would upgrade it.

## 6.4 Limitations

commitment surfaces are gestured at (via passive→active and paraphrase weakness in prior work) but not run at Pythia-410m+.

approximation to true directional patching; a finer, rank-decomposition patch is left to future work.

headline B – A gap suggests. Arm B's absolute patch-CE  $\Delta$  is small (+0.024; Arm C +0.003; B – C only +0.021), and the large B – A gap (+0.758) is mostly produced by Arm A's *negative* patch score (–0.734: ablating "compatibility-aligned" units in a memorizing model *reduces* OOD CE). The E2 patch result therefore supports "B's mechanism differs from A's" strongly, but supports "B localizes a substantial mechanism in the top-k units" only weakly. The large-width sweep sharpens this concern rather than resolving it.

discriminating power at larger widths: a robustness sweep at  $n \in \{17, 19, 23\}$ , hidden width 128, top\_k 16 (`results/e2_e3_neural_larger_n.md`) shows Arm B mean OOD 1.000 vs Arm A 0.089 (gap +0.911) — the OOD story holds decisively — but patch-CE  $\Delta$  for B drops to +0.04 because the trained model spreads the load-bearing structure across more redundant units, so a fixed top-k ablation catches a smaller fraction of the mechanism. A fixed-width top-k patch loses power as width grows; until a rank-normalized or subspace-decomposition patch is run, the E2/E3 localization claim should be read as preliminary. Fix left to future work: normalize patch-CE by hidden-width fraction, or rank-decompose the affected subspace.

hypothesis; the strong form of {allocate, cool, reopen} as the load-bearing subset is only partially tested (Suite C hand/learned/teacher-free re-engagement).

distribution shifts with no clean group action) is open.

confound of Section 6.5: cyclic-orbit augmentation places correctly labeled examples on the held-out deployment support, so E2/E4 do not yet separate *generator learning* from *labeled orbit coverage*.

- Modular addition is one commitment surface; language- and vision-scale
- The patch-CE metric we use in E4 (LoRA-full-ablation) is a coarse
- The E2 patch-CE evidence for *localization* is weaker than the
- The E2/E3 top-k-unit patch-CE metric loses absolute-magnitude
- The anti-Goodhart reading of the correction chain is a compression
- Extension to non-group deployment generators (e.g., semantic
- The highest-priority methodological issue is the label-exposure

## 6.5 Open confound: labeled orbit coverage vs generator learning

E4's compatibility augmentation generates, for each train pair  $(x, y = (x + \text{offset}) \bmod n)$ , the pair  $((x+k) \bmod n, (y+k) \bmod n)$  for random  $k$ . Because  $(y+k) = ((x+k) + \text{offset}) \bmod n$ , every augmented pair is a *correctly labeled* example at input  $(x+k) \bmod n$  — including inputs in the held-out deployment complement. The same holds for E2's  $((a+k) \bmod n, b) \blacksquare (a+b+k) \bmod n$  augmentation. In other words, the intervention arms were trained with direct labeled exposure to the OOD support. The wrong-group Arm C matches augmentation *volume* but places *incorrect* labels on held-out inputs  $(\pi(\text{truth}(x)))$  at input  $\pi(x)$  is generally not  $\text{truth}(\pi(x))$ , so the B-vs-C contrast rules out generic augmentation volume — it does **not** rule out target-support label exposure as the operative mechanism. An earlier revision of Arm C that used correctly-labeled coverage augmentation *did* produce OOD generalization (Appendix A.4, R2), which keeps the coverage explanation live rather than hypothetical.

E4 is therefore consistent with two readings:

generator (cyclic equivariance) and uses it at commitment; the augmentation is how the generator is installed.

OOD orbit with correct labels; nothing transportable is learned beyond fitting the exposed points.

- **Commitment-first (ours)**. The model learns a transportable

- **Coverage (deflationary).** Aligned augmentation simply exposes the

The pre-registered severe follow-up separates them. Train **without ever presenting labels from the held-out deployment orbit**, and compare five arms:

(e.g., an equivariance-consistency loss  $f((x+k) \bmod n)$  vs  $(f(x)+k) \bmod n$  evaluated only where both points are in the train support, or where the target is the model's own prediction rather than a ground-truth label);

(the current Arm B, as the coverage-exposed reference);

structure, matched count of held-out-support exposures);

- aligned generator regularization using only train-support pairs
- aligned augmentation that is allowed to enter held-out support
- wrong-generator regularization (matched form of (1), wrong group);
- ordinary coverage-matched augmentation (correct labels, no group)
- readout selection (current Arm A).

Evaluate on a **new group element or modulus not used by the intervention**, with rank-normalized causal patching.

**Kill criteria (pre-registered).** The commitment-first interpretation is materially weakened if any of:

deployment-support points (i.e.,  $(1) \approx (5)$  while  $(2) \gg (5)$ );

augmentation (2);

- arm (1)'s gains disappear when augmentation cannot label
- coverage-matched augmentation (4) performs on par with aligned
- patch-CE fails to predict transfer to a novel commitment surface.

Until this follow-up runs, the E4 claim should be read as: *train-time aligned intervention recovers external OOD where readout selection does not* — with the mechanism (transportable generator vs labeled coverage) not yet isolated.

## 7. Conclusion

Availability is not use; probe AUC is not causal path; portable weakness is not portable simplicity. Structure-in-representation is a starting point, not the discovery. The discovery — the thing you would defend to a skeptical reviewer three moves out — is *commitment-surface survival*: did the causal effect of the identified mechanism survive transport, gauge-fix, and change of commitment on a system you did not build? If yes, the geometry story becomes an aligned-generator special case with clean group theory. If no, the geometry story was a footprint of the prior. Both are useful, but only the first is science.

## 8. Author's stance

Attribution: human director Jawaun Brown. Code, sweeps, and paper drafts are agent-generated under human direction and review; results are committed with per-cell JSON provenance and pre-registered gates. Prior program overreach (portable scalar concern, weakness-as-universal-law readings) is corrected here in the direction indicated by the anomaly map in ``notes/weakness_topology_program_synthesis.md` and the pre-registered kills in `docs/external_contact_preregistration.md`.

## Appendix

### A.1 Full derivations for Props 1 and 2

**Prop. 1 (non-identification of causal use by probe AUC).** Let  $M_\alpha$  be the set of admissible hypotheses. Fix a probe  $p : X \rightarrow [0, 1]$  and any admissible baseline  $f'$  with a known load-bearing mechanism  $M(f')$  such that  $CE(f' \mid \text{patch}, C) \geq \epsilon$  when the patch zeroes  $M(f')$ . We construct  $f$  as follows: (i) duplicate  $f'$ 's decision head onto a fresh subspace  $S$  orthogonal to  $M(f')$  in feature space; (ii) copy  $f'$ 's decisions bit-for-bit onto  $M(f') \oplus S$ ; (iii) route the probe input through  $S$  while keeping the downstream computation routed through  $M(f')$ . By construction:

Property (a).  $\text{AUC}(p \cdot f) = \text{AUC}(p \cdot f')$  on any input distribution, because  $p$  reads  $S$  and  $S$  was constructed to match  $f'$ 's probe response. Property (b). Zero-ablating  $S$  changes  $p \cdot f$  but not the decision head's output, so  $\text{CE}(f \mid \text{patch}(S), C) = 0$ . Property (c). Zero-ablating  $M(f')$  changes both the decision output and the probe response, so  $\text{CE}(f \mid \text{patch}(M(f')), C) = \text{CE}(f' \mid \text{patch}, C) \geq \epsilon$ .

This exhibits two admissible hypotheses with identical probe AUC and causal-effect gap  $\epsilon$  under the respective identified-mechanism patches. To realize an arbitrary intermediate gap  $\delta \in [0, \epsilon]$ , interpolate the *routing*, not the functions: construct  $f_\lambda$  that routes a fraction  $\lambda$  of the decision head's input mass through  $M(f')$  and  $1 - \lambda$  through the (causally inert, probe-visible) copy on  $S$ , with the head re-normalized so the input-output behavior — hence admissibility and probe response — is unchanged.  $\text{AUC}(p \cdot f_\lambda)$  is constant in  $\lambda$  while  $\text{CE}(f_\lambda \mid \text{patch}(S), C)$  moves continuously from 0 (at  $\lambda = 1$ ) toward  $\epsilon$  (at  $\lambda = 0$ ), so every gap  $\delta \in [0, \epsilon]$  is attained at equal AUC.

Hence no function of probe AUC alone identifies, or lower-bounds,  $\text{CE}(\cdot \mid \text{patch}, C)$  on  $M_\alpha$ . We emphasize the scope: this is a *non-identification* theorem over the admissible class. It does **not** assert probabilistic independence of AUC and CE — that would require a distribution over  $M_\alpha$ , and under particular training distributions the two can correlate (E3 measures such a correlation in the aligned regime). What fails is the inference from probe evidence to causal use for any *individual* hypothesis. ■

**Prop. 2 (weakness is diagnostic exactly when aligned).** *Deployment model.* Both groups finite. A deployment slice is generated by drawing  $g \sim \text{Uniform}(G_{\text{dep}})$  (the prior) and evaluating the candidate on the  $g$ -translated support; the candidate scores 1 on the slice iff it is compatible with  $g$  (the orbit-likelihood), and concern is uniform. Under this model the expected deployment accuracy of an admissible  $f$  is

$$E_g[\text{acc}(f)] = W_{\{G_{\text{dep}}\}}(f) / |G_{\text{dep}}|,$$

i.e. expected accuracy is *exactly* normalized deployment-restricted weakness.

(i) *Aligned case.* If the selector scores candidates by  $W_{\{G_{\text{dep}}\}}$  — either because  $G_{\text{probe}} = G_{\text{dep}}$ , or because weakness is explicitly restricted (or concern-weighted) to  $G_{\text{dep}}$  — then  $\arg\max_f W_{\{G_{\text{dep}}\}}(f)$  maximizes  $E_g[\text{acc}(f)]$  by the identity above. Since the score is a deterministic function of  $f$  and the objective is its own posterior expectation under the stated prior and likelihood, the argmax selector is Bayes-optimal among selectors that depend only on  $f$ . (Bayes-optimality here is *relative to this explicit prior and likelihood*; no claim is made under other priors.)

(ii) *Superset counterexample.*  $G_{\text{probe}} \supset G_{\text{dep}}$  does not suffice. Let  $G_{\text{dep}} = \{e, g\}$  and  $G_{\text{probe}} = \{e, g, h_1, h_2, h_3\}$ . Take  $f$  compatible with  $\{e, h_1, h_2, h_3\}$  (so  $W_{\{G_{\text{probe}}\}}(f) = 4, W_{\{G_{\text{dep}}\}}(f) = 1$ ) and  $f'$  compatible with  $\{e, g\}$  (so  $W_{\{G_{\text{probe}}\}}(f') = 2, W_{\{G_{\text{dep}}\}}(f') = 2$ ). Then  $W_{\{G_{\text{probe}}\}}(f) > W_{\{G_{\text{probe}}\}}(f')$  while  $E_g[\text{acc}(f)] = 1/2 < 1 = E_g[\text{acc}(f')]$ : the superset-probe ranking is inverted on deployment. Optimality from a superset probe group is recovered only under an ordering-preservation assumption on the candidate set:  $W_{\{G_{\text{probe}}\}}(f) > W_{\{G_{\text{probe}}\}}(f') \Rightarrow W_{\{G_{\text{dep}}\}}(f) \geq W_{\{G_{\text{dep}}\}}(f')$ .

(iii) *Non-aligned case.* Suppose  $G_{\text{dep}} \not\subset G_{\text{probe}}$ . Let  $g' \in G_{\text{dep}} \setminus G_{\text{probe}}$ . Then  $W_{\{G_{\text{probe}}\}}(f)$  does not measure compatibility with  $g'$ . Construct  $f, f'$  admissible such that  $W_{\{G_{\text{probe}}\}}(f) > W_{\{G_{\text{probe}}\}}(f')$  but  $f'$  is compatible with  $g'$  and  $f$  is not; then  $f'$  has strictly higher expected deployment accuracy on the  $g'$ -generated slice while  $f$  has higher  $W_{\{G_{\text{probe}}\}}$  — a negative association between selector score and OOD accuracy on that candidate pair. Hence  $W_{\{G_{\text{probe}}\}}$  is a footprint whose sign of association with deployment accuracy is generator-dependent. ■

**Corollary (order-equivalence for concern-weighted extension mass).** Let  $C : U \rightarrow \mathbb{R}_{\geq 0}$  be a concern measure on the deployment slice  $U$ ,  $C_{\text{star}}$  the true consequence measure, and  $F \subset M_\alpha$  a finite candidate set. *Sufficiency:* if  $C = a \cdot C_{\text{star}}$  for some  $a > 0$ , then  $W_C = a \cdot W_{\{C_{\text{star}}\}}$  and  $\arg\max_F W_C = \arg\max_F W_{\{C_{\text{star}}\}}$ , so the selector is  $C_{\text{star}}$ -optimal on  $F$ . *Exact condition:* the selector  $\arg\max_F W_C$  picks a  $C_{\text{star}}$ -optimal candidate for every truth assignment iff  $C$  is order-equivalent to  $C_{\text{star}}$  on  $F$ :

$$W_C(f, U) > W_C(f', U) \Leftrightarrow W_{\{C_{\text{star}}\}}(f, U) > W_{\{C_{\text{star}}\}}(f', U) \text{ for all } f, f' \in F.$$

Equality up to positive scale is *not necessary*: over a finite  $F$ , distinct weightings that induce the same ranking of candidates are equally optimal, so the earlier "iff  $C = C_{\text{star}}$  up to scaling" phrasing was too strong and is retracted here in favor of the order-equivalence condition. *Misspecification:* if  $C$  is a random assignment with the same marginal distribution as  $C_{\text{star}}$ , independent of candidate coverage, then  $E[W_C(f, U)] = E[C] \cdot |\{x \in U : f(x) = \text{truth}(x)\}|$  — proportional to unweighted extension mass — so the misspecified selector reduces to unweighted Bennett *in expectation* (any particular draw can be mildly helpful or mildly adversarial; E1's realized draw was mildly adversarial at  $-0.054$ ). A strictly adversarial  $C$  (anti-correlated with  $C_{\text{star}}$ ) inverts the relation. ■

## A.2 Per-cell tables

The PDF build renders complete per-cell tables directly from the committed public-safe E1 and E2/E3 JSON summaries and the compact committed E4 appendix artifact. The E2 and E3 analyses share one 216-cell population. E4 exports all 108 cells while intentionally omitting raw function tables, train/OOD input lists, and parameter metadata; no requested appendix metric is unavailable.

**A.2.1 E1.** All 96 concern-weighted selector cells from `experiments/commitment_surface/results/e1_concern_weighted.json`.

| n  | Seed | Unweighted | Well-spec | Misspec | Loss  | Truth |
|----|------|------------|-----------|---------|-------|-------|
| 7  | 0    | 0.591      | 0.848     | 0.515   | 0.197 | 1.000 |
| 7  | 1    | 0.621      | 0.864     | 0.561   | 0.364 | 1.000 |
| 7  | 2    | 0.879      | 0.879     | 0.545   | 0.409 | 1.000 |
| 7  | 3    | 0.470      | 0.864     | 0.409   | 0.045 | 1.000 |
| 7  | 4    | 0.591      | 0.848     | 0.561   | 0.591 | 1.000 |
| 7  | 5    | 0.606      | 0.879     | 0.606   | 0.076 | 1.000 |
| 7  | 6    | 0.621      | 0.848     | 0.455   | 0.061 | 1.000 |
| 7  | 7    | 0.318      | 0.848     | 0.318   | 0.379 | 1.000 |
| 7  | 8    | 0.591      | 0.864     | 0.576   | 0.424 | 1.000 |
| 7  | 9    | 0.864      | 0.864     | 0.439   | 0.500 | 1.000 |
| 7  | 10   | 0.591      | 0.818     | 0.727   | 0.500 | 1.000 |
| 7  | 11   | 0.606      | 0.879     | 0.682   | 0.076 | 1.000 |
| 7  | 12   | 0.742      | 0.848     | 0.455   | 0.364 | 1.000 |
| 7  | 13   | 0.606      | 0.864     | 0.697   | 0.364 | 1.000 |
| 7  | 14   | 0.591      | 0.833     | 0.288   | 0.394 | 1.000 |
| 7  | 15   | 0.591      | 0.864     | 0.303   | 0.227 | 1.000 |
| 7  | 16   | 0.621      | 0.879     | 0.712   | 0.288 | 1.000 |
| 7  | 17   | 0.742      | 0.864     | 0.394   | 0.227 | 1.000 |
| 7  | 18   | 0.879      | 0.879     | 0.727   | 0.545 | 1.000 |
| 7  | 19   | 0.439      | 0.848     | 0.682   | 0.227 | 1.000 |
| 7  | 20   | 0.333      | 0.848     | 0.333   | 0.545 | 1.000 |
| 7  | 21   | 0.455      | 0.848     | 0.424   | 0.227 | 1.000 |
| 7  | 22   | 0.879      | 0.879     | 0.879   | 0.212 | 1.000 |
| 7  | 23   | 0.758      | 0.833     | 0.576   | 0.212 | 1.000 |
| 7  | 24   | 0.606      | 0.864     | 0.712   | 0.212 | 1.000 |
| 7  | 25   | 0.712      | 0.848     | 0.712   | 0.045 | 1.000 |
| 7  | 26   | 0.742      | 0.864     | 0.591   | 0.242 | 1.000 |
| 7  | 27   | 0.606      | 0.833     | 0.530   | 0.485 | 1.000 |
| 7  | 28   | 0.182      | 0.833     | 0.561   | 0.561 | 1.000 |
| 7  | 29   | 0.727      | 0.848     | 0.727   | 0.182 | 1.000 |
| 7  | 30   | 0.742      | 0.879     | 0.409   | 0.227 | 1.000 |
| 7  | 31   | 0.591      | 0.848     | 0.409   | 0.273 | 1.000 |
| 11 | 0    | 0.547      | 0.796     | 0.619   | 0.238 | 1.000 |
| 11 | 1    | 0.564      | 0.829     | 0.564   | 0.271 | 1.000 |
| 11 | 2    | 0.862      | 0.862     | 0.862   | 0.271 | 1.000 |
| 11 | 3    | 0.646      | 0.779     | 0.497   | 0.309 | 1.000 |
| 11 | 4    | 0.398      | 0.812     | 0.381   | 0.453 | 1.000 |
| 11 | 5    | 0.790      | 0.823     | 0.779   | 0.177 | 1.000 |
| 11 | 6    | 0.503      | 0.773     | 0.326   | 0.370 | 1.000 |
| 11 | 7    | 0.459      | 0.823     | 0.381   | 0.359 | 1.000 |
| 11 | 8    | 0.448      | 0.840     | 0.779   | 0.381 | 1.000 |
| 11 | 9    | 0.343      | 0.773     | 0.630   | 0.448 | 1.000 |
| 11 | 10   | 0.746      | 0.818     | 0.619   | 0.448 | 1.000 |
| 11 | 11   | 0.547      | 0.807     | 0.497   | 0.315 | 1.000 |
| 11 | 12   | 0.547      | 0.818     | 0.370   | 0.293 | 1.000 |
| 11 | 13   | 0.602      | 0.818     | 0.387   | 0.232 | 1.000 |
| 11 | 14   | 0.796      | 0.829     | 0.381   | 0.304 | 1.000 |
| 11 | 15   | 0.796      | 0.796     | 0.718   | 0.293 | 1.000 |
| 11 | 16   | 0.602      | 0.785     | 0.602   | 0.409 | 1.000 |
| 11 | 17   | 0.442      | 0.823     | 0.453   | 0.199 | 1.000 |
| 11 | 18   | 0.696      | 0.840     | 0.470   | 0.155 | 1.000 |
| 11 | 19   | 0.602      | 0.790     | 0.271   | 0.343 | 1.000 |
| 11 | 20   | 0.547      | 0.829     | 0.431   | 0.420 | 1.000 |

| n  | Seed | Unweighted | Well-spec | Misspec | Loss  | Truth |
|----|------|------------|-----------|---------|-------|-------|
| 11 | 21   | 0.453      | 0.829     | 0.326   | 0.403 | 1.000 |
| 11 | 22   | 0.840      | 0.840     | 0.834   | 0.182 | 1.000 |
| 11 | 23   | 0.287      | 0.807     | 0.271   | 0.354 | 1.000 |
| 11 | 24   | 0.381      | 0.762     | 0.680   | 0.260 | 1.000 |
| 11 | 25   | 0.696      | 0.823     | 0.442   | 0.166 | 1.000 |
| 11 | 26   | 0.608      | 0.773     | 0.392   | 0.304 | 1.000 |
| 11 | 27   | 0.564      | 0.845     | 0.530   | 0.514 | 1.000 |
| 11 | 28   | 0.442      | 0.812     | 0.724   | 0.315 | 1.000 |
| 11 | 29   | 0.602      | 0.829     | 0.376   | 0.320 | 1.000 |
| 11 | 30   | 0.597      | 0.829     | 0.376   | 0.199 | 1.000 |
| 11 | 31   | 0.387      | 0.812     | 0.238   | 0.414 | 1.000 |
| 13 | 0    | 0.409      | 0.795     | 0.735   | 0.201 | 1.000 |
| 13 | 1    | 0.493      | 0.752     | 0.493   | 0.292 | 1.000 |
| 13 | 2    | 0.406      | 0.752     | 0.735   | 0.326 | 1.000 |
| 13 | 3    | 0.436      | 0.789     | 0.352   | 0.144 | 1.000 |
| 13 | 4    | 0.440      | 0.765     | 0.426   | 0.430 | 1.000 |
| 13 | 5    | 0.678      | 0.779     | 0.634   | 0.302 | 1.000 |
| 13 | 6    | 0.403      | 0.785     | 0.295   | 0.174 | 1.000 |
| 13 | 7    | 0.477      | 0.738     | 0.477   | 0.349 | 1.000 |
| 13 | 8    | 0.752      | 0.755     | 0.406   | 0.470 | 1.000 |
| 13 | 9    | 0.648      | 0.748     | 0.480   | 0.302 | 1.000 |
| 13 | 10   | 0.537      | 0.805     | 0.218   | 0.268 | 1.000 |
| 13 | 11   | 0.346      | 0.792     | 0.792   | 0.262 | 1.000 |
| 13 | 12   | 0.312      | 0.758     | 0.362   | 0.326 | 1.000 |
| 13 | 13   | 0.409      | 0.822     | 0.376   | 0.268 | 1.000 |
| 13 | 14   | 0.560      | 0.795     | 0.795   | 0.416 | 1.000 |
| 13 | 15   | 0.678      | 0.789     | 0.463   | 0.292 | 1.000 |
| 13 | 16   | 0.470      | 0.762     | 0.436   | 0.493 | 1.000 |
| 13 | 17   | 0.688      | 0.779     | 0.349   | 0.349 | 1.000 |
| 13 | 18   | 0.463      | 0.779     | 0.255   | 0.205 | 1.000 |
| 13 | 19   | 0.738      | 0.768     | 0.661   | 0.295 | 1.000 |
| 13 | 20   | 0.792      | 0.792     | 0.735   | 0.315 | 1.000 |
| 13 | 21   | 0.450      | 0.779     | 0.450   | 0.272 | 1.000 |
| 13 | 22   | 0.356      | 0.775     | 0.671   | 0.302 | 1.000 |
| 13 | 23   | 0.685      | 0.728     | 0.594   | 0.406 | 1.000 |
| 13 | 24   | 0.386      | 0.792     | 0.346   | 0.265 | 1.000 |
| 13 | 25   | 0.725      | 0.762     | 0.725   | 0.262 | 1.000 |
| 13 | 26   | 0.735      | 0.762     | 0.557   | 0.443 | 1.000 |
| 13 | 27   | 0.285      | 0.738     | 0.315   | 0.258 | 1.000 |
| 13 | 28   | 0.473      | 0.762     | 0.436   | 0.309 | 1.000 |
| 13 | 29   | 0.386      | 0.799     | 0.406   | 0.285 | 1.000 |
| 13 | 30   | 0.466      | 0.775     | 0.416   | 0.342 | 1.000 |
| 13 | 31   | 0.419      | 0.772     | 0.419   | 0.295 | 1.000 |

**A.2.2 E2/E3.** All 216 trained neural cells. E2 arm comparisons and E3 correlations use this same committed cell population from experiments/commitment\_surface/results/e2\_e3\_neural.json.

| Arm | n | Seed     | Train frac | OOD   | Patch-CE | Wrong patch | Weakness |
|-----|---|----------|------------|-------|----------|-------------|----------|
| A   | 7 | 20261080 | 0.40       | 0.000 | -0.583   | -0.649      | 0.143    |
| B   | 7 | 20261080 | 0.40       | 1.000 | 0.051    | 0.002       | 1.000    |
| C   | 7 | 20261080 | 0.40       | 0.172 | 0.000    | 0.000       | 0.143    |
| D   | 7 | 20261080 | 0.40       | 0.000 | -0.583   | -0.649      | 0.143    |
| A   | 7 | 20261181 | 0.40       | 0.000 | 0.000    | 0.000       | 0.143    |
| B   | 7 | 20261181 | 0.40       | 1.000 | 0.027    | 0.002       | 1.000    |
| C   | 7 | 20261181 | 0.40       | 0.138 | 0.000    | 0.000       | 0.143    |
| D   | 7 | 20261181 | 0.40       | 0.000 | 0.000    | 0.000       | 0.143    |
| A   | 7 | 20261282 | 0.40       | 0.000 | 0.000    | 0.000       | 0.143    |
| B   | 7 | 20261282 | 0.40       | 1.000 | 0.015    | 0.001       | 1.000    |
| C   | 7 | 20261282 | 0.40       | 0.138 | 0.000    | 0.000       | 0.143    |
| D   | 7 | 20261282 | 0.40       | 0.000 | 0.000    | 0.000       | 0.143    |
| A   | 7 | 20261383 | 0.40       | 0.000 | -0.138   | -0.338      | 0.143    |
| B   | 7 | 20261383 | 0.40       | 1.000 | 0.029    | 0.005       | 1.000    |
| C   | 7 | 20261383 | 0.40       | 0.310 | 0.000    | 0.000       | 0.143    |
| D   | 7 | 20261383 | 0.40       | 0.000 | -0.138   | -0.338      | 0.143    |
| A   | 7 | 20261484 | 0.40       | 0.000 | -0.212   | -0.438      | 0.143    |
| B   | 7 | 20261484 | 0.40       | 0.759 | -0.165   | -0.096      | 0.143    |
| C   | 7 | 20261484 | 0.40       | 0.103 | 0.000    | 0.000       | 0.143    |
| D   | 7 | 20261484 | 0.40       | 0.000 | -0.212   | -0.438      | 0.143    |
| A   | 7 | 20261585 | 0.40       | 0.000 | -0.323   | -0.274      | 0.143    |
| B   | 7 | 20261585 | 0.40       | 1.000 | 0.006    | 0.002       | 1.000    |
| C   | 7 | 20261585 | 0.40       | 0.310 | 0.000    | 0.000       | 0.143    |
| D   | 7 | 20261585 | 0.40       | 0.000 | -0.323   | -0.274      | 0.143    |
| A   | 7 | 20261185 | 0.55       | 0.000 | -1.453   | -0.659      | 0.143    |
| B   | 7 | 20261185 | 0.55       | 1.000 | 0.028    | 0.002       | 1.000    |
| C   | 7 | 20261185 | 0.55       | 0.318 | 0.000    | 0.000       | 0.143    |
| D   | 7 | 20261185 | 0.55       | 0.000 | -1.453   | -0.659      | 0.143    |
| A   | 7 | 20261286 | 0.55       | 0.000 | 0.191    | -0.254      | 0.143    |
| B   | 7 | 20261286 | 0.55       | 1.000 | 0.022    | 0.001       | 1.000    |
| C   | 7 | 20261286 | 0.55       | 0.273 | -0.006   | 0.013       | 0.143    |
| D   | 7 | 20261286 | 0.55       | 0.000 | 0.191    | -0.254      | 0.143    |
| A   | 7 | 20261387 | 0.55       | 0.000 | -0.161   | -0.313      | 0.143    |
| B   | 7 | 20261387 | 0.55       | 1.000 | 0.026    | 0.007       | 1.000    |
| C   | 7 | 20261387 | 0.55       | 0.182 | 0.013    | 0.025       | 0.143    |
| D   | 7 | 20261387 | 0.55       | 0.000 | -0.161   | -0.313      | 0.143    |
| A   | 7 | 20261488 | 0.55       | 0.000 | -0.859   | -0.469      | 0.143    |
| B   | 7 | 20261488 | 0.55       | 1.000 | 0.029    | 0.001       | 1.000    |
| C   | 7 | 20261488 | 0.55       | 0.227 | 0.022    | 0.002       | 0.143    |
| D   | 7 | 20261488 | 0.55       | 0.000 | -0.859   | -0.469      | 0.143    |
| A   | 7 | 20261589 | 0.55       | 0.000 | 0.000    | 0.000       | 0.143    |
| B   | 7 | 20261589 | 0.55       | 1.000 | 0.032    | 0.001       | 1.000    |
| C   | 7 | 20261589 | 0.55       | 0.182 | 0.000    | 0.000       | 0.143    |
| D   | 7 | 20261589 | 0.55       | 0.000 | 0.000    | 0.000       | 0.143    |
| A   | 7 | 20261690 | 0.55       | 0.000 | -0.160   | -0.717      | 0.143    |
| B   | 7 | 20261690 | 0.55       | 1.000 | 0.004    | 0.001       | 1.000    |
| C   | 7 | 20261690 | 0.55       | 0.318 | -0.001   | 0.001       | 0.143    |
| D   | 7 | 20261690 | 0.55       | 0.000 | -0.160   | -0.717      | 0.143    |
| A   | 7 | 20261290 | 0.70       | 0.000 | -1.012   | -1.279      | 0.143    |
| B   | 7 | 20261290 | 0.70       | 1.000 | 0.023    | 0.003       | 1.000    |
| C   | 7 | 20261290 | 0.70       | 0.067 | -0.004   | -0.004      | 0.143    |
| D   | 7 | 20261290 | 0.70       | 0.000 | -1.012   | -1.279      | 0.143    |
| A   | 7 | 20261391 | 0.70       | 0.000 | -0.481   | -1.114      | 0.143    |
| B   | 7 | 20261391 | 0.70       | 1.000 | 0.014    | 0.008       | 1.000    |
| C   | 7 | 20261391 | 0.70       | 0.200 | -0.006   | -0.012      | 0.143    |
| D   | 7 | 20261391 | 0.70       | 0.000 | -0.481   | -1.114      | 0.143    |
| A   | 7 | 20261492 | 0.70       | 0.000 | -1.182   | -1.427      | 0.143    |
| B   | 7 | 20261492 | 0.70       | 1.000 | 0.014    | 0.001       | 1.000    |



| Arm | n  | Seed     | Train frac | OOD   | Patch-CE | Wrong patch | Weakness |
|-----|----|----------|------------|-------|----------|-------------|----------|
| C   | 7  | 20261492 | 0.70       | 0.200 | -0.007   | 0.008       | 0.143    |
| D   | 7  | 20261492 | 0.70       | 0.000 | -1.182   | -1.427      | 0.143    |
| A   | 7  | 20261593 | 0.70       | 0.000 | -0.421   | -0.857      | 0.143    |
| B   | 7  | 20261593 | 0.70       | 1.000 | 0.028    | 0.002       | 1.000    |
| C   | 7  | 20261593 | 0.70       | 0.067 | -0.008   | -0.006      | 0.143    |
| D   | 7  | 20261593 | 0.70       | 0.000 | -0.421   | -0.857      | 0.143    |
| A   | 7  | 20261694 | 0.70       | 0.000 | 0.058    | -1.453      | 0.143    |
| B   | 7  | 20261694 | 0.70       | 1.000 | 0.016    | 0.001       | 1.000    |
| C   | 7  | 20261694 | 0.70       | 0.200 | -0.001   | -0.019      | 0.143    |
| D   | 7  | 20261694 | 0.70       | 0.000 | 0.058    | -1.453      | 0.143    |
| A   | 7  | 20261795 | 0.70       | 0.000 | -0.580   | -0.541      | 0.143    |
| B   | 7  | 20261795 | 0.70       | 1.000 | 0.070    | 0.002       | 1.000    |
| C   | 7  | 20261795 | 0.70       | 0.200 | -0.010   | 0.016       | 0.143    |
| D   | 7  | 20261795 | 0.70       | 0.000 | -0.580   | -0.541      | 0.143    |
| A   | 11 | 20261132 | 0.40       | 0.000 | -0.906   | -0.617      | 0.091    |
| B   | 11 | 20261132 | 0.40       | 1.000 | 0.015    | 0.008       | 1.000    |
| C   | 11 | 20261132 | 0.40       | 0.164 | -0.003   | 0.014       | 0.091    |
| D   | 11 | 20261132 | 0.40       | 0.000 | -0.906   | -0.617      | 0.091    |
| A   | 11 | 20261233 | 0.40       | 0.000 | -0.748   | -0.707      | 0.091    |
| B   | 11 | 20261233 | 0.40       | 1.000 | 0.048    | 0.003       | 1.000    |
| C   | 11 | 20261233 | 0.40       | 0.151 | 0.000    | 0.007       | 0.091    |
| D   | 11 | 20261233 | 0.40       | 0.000 | -0.748   | -0.707      | 0.091    |
| A   | 11 | 20261334 | 0.40       | 0.000 | -0.328   | -0.346      | 0.091    |
| B   | 11 | 20261334 | 0.40       | 1.000 | 0.013    | 0.002       | 1.000    |
| C   | 11 | 20261334 | 0.40       | 0.151 | 0.000    | 0.000       | 0.091    |
| D   | 11 | 20261334 | 0.40       | 0.000 | -0.328   | -0.346      | 0.091    |
| A   | 11 | 20261435 | 0.40       | 0.000 | -0.799   | -1.100      | 0.091    |
| B   | 11 | 20261435 | 0.40       | 1.000 | 0.029    | 0.008       | 1.000    |
| C   | 11 | 20261435 | 0.40       | 0.151 | 0.008    | 0.011       | 0.091    |
| D   | 11 | 20261435 | 0.40       | 0.000 | -0.799   | -1.100      | 0.091    |
| A   | 11 | 20261536 | 0.40       | 0.000 | -0.160   | -0.941      | 0.091    |
| B   | 11 | 20261536 | 0.40       | 1.000 | 0.016    | 0.005       | 1.000    |
| C   | 11 | 20261536 | 0.40       | 0.164 | 0.006    | 0.010       | 0.091    |
| D   | 11 | 20261536 | 0.40       | 0.000 | -0.160   | -0.941      | 0.091    |
| A   | 11 | 20261637 | 0.40       | 0.000 | -0.534   | -0.985      | 0.091    |
| B   | 11 | 20261637 | 0.40       | 1.000 | 0.005    | 0.004       | 1.000    |
| C   | 11 | 20261637 | 0.40       | 0.151 | 0.008    | 0.029       | 0.091    |
| D   | 11 | 20261637 | 0.40       | 0.000 | -0.534   | -0.985      | 0.091    |
| A   | 11 | 20261237 | 0.55       | 0.000 | -0.961   | -0.678      | 0.091    |
| B   | 11 | 20261237 | 0.55       | 1.000 | 0.058    | 0.003       | 1.000    |
| C   | 11 | 20261237 | 0.55       | 0.204 | -0.001   | 0.007       | 0.091    |
| D   | 11 | 20261237 | 0.55       | 0.000 | -0.961   | -0.678      | 0.091    |
| A   | 11 | 20261338 | 0.55       | 0.000 | -0.004   | -1.338      | 0.091    |
| B   | 11 | 20261338 | 0.55       | 1.000 | 0.033    | 0.005       | 1.000    |
| C   | 11 | 20261338 | 0.55       | 0.074 | 0.001    | 0.003       | 0.091    |
| D   | 11 | 20261338 | 0.55       | 0.000 | -0.004   | -1.338      | 0.091    |
| A   | 11 | 20261439 | 0.55       | 0.000 | -0.307   | -0.327      | 0.091    |
| B   | 11 | 20261439 | 0.55       | 1.000 | 0.006    | 0.001       | 1.000    |
| C   | 11 | 20261439 | 0.55       | 0.148 | -0.004   | 0.022       | 0.091    |
| D   | 11 | 20261439 | 0.55       | 0.000 | -0.307   | -0.327      | 0.091    |
| A   | 11 | 20261540 | 0.55       | 0.000 | -0.881   | -0.980      | 0.091    |
| B   | 11 | 20261540 | 0.55       | 1.000 | 0.011    | 0.006       | 1.000    |
| C   | 11 | 20261540 | 0.55       | 0.130 | 0.015    | 0.027       | 0.091    |
| D   | 11 | 20261540 | 0.55       | 0.000 | -0.881   | -0.980      | 0.091    |
| A   | 11 | 20261641 | 0.55       | 0.000 | -0.840   | -0.920      | 0.091    |
| B   | 11 | 20261641 | 0.55       | 1.000 | 0.074    | 0.003       | 1.000    |
| C   | 11 | 20261641 | 0.55       | 0.204 | 0.009    | -0.020      | 0.091    |
| D   | 11 | 20261641 | 0.55       | 0.000 | -0.840   | -0.920      | 0.091    |
| A   | 11 | 20261742 | 0.55       | 0.000 | -0.844   | -0.813      | 0.091    |
| B   | 11 | 20261742 | 0.55       | 1.000 | 0.039    | 0.007       | 1.000    |
| C   | 11 | 20261742 | 0.55       | 0.148 | -0.006   | 0.013       | 0.091    |

| Arm | n  | Seed     | Train frac | OOD   | Patch-CE | Wrong patch | Weakness |
|-----|----|----------|------------|-------|----------|-------------|----------|
| D   | 11 | 20261742 | 0.55       | 0.000 | -0.844   | -0.813      | 0.091    |
| A   | 11 | 20261342 | 0.70       | 0.000 | -0.623   | -0.854      | 0.091    |
| B   | 11 | 20261342 | 0.70       | 1.000 | 0.022    | 0.010       | 1.000    |
| C   | 11 | 20261342 | 0.70       | 0.167 | 0.003    | 0.006       | 0.091    |
| D   | 11 | 20261342 | 0.70       | 0.000 | -0.623   | -0.854      | 0.091    |
| A   | 11 | 20261443 | 0.70       | 0.000 | -0.519   | -1.052      | 0.091    |
| B   | 11 | 20261443 | 0.70       | 1.000 | 0.008    | 0.006       | 1.000    |
| C   | 11 | 20261443 | 0.70       | 0.083 | 0.023    | 0.017       | 0.091    |
| D   | 11 | 20261443 | 0.70       | 0.000 | -0.519   | -1.052      | 0.091    |
| A   | 11 | 20261544 | 0.70       | 0.000 | -0.346   | -0.801      | 0.091    |
| B   | 11 | 20261544 | 0.70       | 1.000 | 0.019    | 0.002       | 1.000    |
| C   | 11 | 20261544 | 0.70       | 0.139 | 0.040    | 0.042       | 0.091    |
| D   | 11 | 20261544 | 0.70       | 0.000 | -0.346   | -0.801      | 0.091    |
| A   | 11 | 20261645 | 0.70       | 0.000 | -0.401   | -0.687      | 0.091    |
| B   | 11 | 20261645 | 0.70       | 1.000 | 0.137    | 0.003       | 1.000    |
| C   | 11 | 20261645 | 0.70       | 0.250 | 0.029    | 0.028       | 0.091    |
| D   | 11 | 20261645 | 0.70       | 0.000 | -0.401   | -0.687      | 0.091    |
| A   | 11 | 20261746 | 0.70       | 0.000 | -0.535   | -0.807      | 0.091    |
| B   | 11 | 20261746 | 0.70       | 1.000 | 0.042    | 0.002       | 1.000    |
| C   | 11 | 20261746 | 0.70       | 0.250 | 0.014    | 0.002       | 0.091    |
| D   | 11 | 20261746 | 0.70       | 0.000 | -0.535   | -0.807      | 0.091    |
| A   | 11 | 20261847 | 0.70       | 0.000 | -0.444   | -0.903      | 0.091    |
| B   | 11 | 20261847 | 0.70       | 1.000 | 0.004    | 0.002       | 1.000    |
| C   | 11 | 20261847 | 0.70       | 0.056 | -0.015   | 0.034       | 0.091    |
| D   | 11 | 20261847 | 0.70       | 0.000 | -0.444   | -0.903      | 0.091    |
| A   | 13 | 20261158 | 0.40       | 0.000 | -0.438   | -0.527      | 0.077    |
| B   | 13 | 20261158 | 0.40       | 0.871 | -0.055   | -0.030      | 0.077    |
| C   | 13 | 20261158 | 0.40       | 0.119 | -0.000   | 0.024       | 0.077    |
| D   | 13 | 20261158 | 0.40       | 0.000 | -0.438   | -0.527      | 0.077    |
| A   | 13 | 20261259 | 0.40       | 0.000 | -0.733   | -0.365      | 0.077    |
| B   | 13 | 20261259 | 0.40       | 1.000 | 0.016    | 0.010       | 1.000    |
| C   | 13 | 20261259 | 0.40       | 0.149 | 0.000    | -0.002      | 0.077    |
| D   | 13 | 20261259 | 0.40       | 0.000 | -0.733   | -0.365      | 0.077    |
| A   | 13 | 20261360 | 0.40       | 0.000 | -0.187   | -0.586      | 0.077    |
| B   | 13 | 20261360 | 0.40       | 1.000 | 0.043    | 0.009       | 1.000    |
| C   | 13 | 20261360 | 0.40       | 0.129 | 0.007    | 0.009       | 0.077    |
| D   | 13 | 20261360 | 0.40       | 0.000 | -0.187   | -0.586      | 0.077    |
| A   | 13 | 20261461 | 0.40       | 0.000 | -1.106   | -0.529      | 0.077    |
| B   | 13 | 20261461 | 0.40       | 1.000 | 0.026    | 0.006       | 1.000    |
| C   | 13 | 20261461 | 0.40       | 0.149 | 0.000    | 0.000       | 0.077    |
| D   | 13 | 20261461 | 0.40       | 0.000 | -1.106   | -0.529      | 0.077    |
| A   | 13 | 20261562 | 0.40       | 0.000 | -0.156   | -0.597      | 0.077    |
| B   | 13 | 20261562 | 0.40       | 1.000 | 0.014    | 0.004       | 1.000    |
| C   | 13 | 20261562 | 0.40       | 0.109 | -0.004   | -0.000      | 0.077    |
| D   | 13 | 20261562 | 0.40       | 0.000 | -0.156   | -0.597      | 0.077    |
| A   | 13 | 20261663 | 0.40       | 0.000 | -1.055   | -0.989      | 0.077    |
| B   | 13 | 20261663 | 0.40       | 1.000 | 0.056    | 0.003       | 1.000    |
| C   | 13 | 20261663 | 0.40       | 0.149 | 0.002    | 0.025       | 0.077    |
| D   | 13 | 20261663 | 0.40       | 0.000 | -1.055   | -0.989      | 0.077    |
| A   | 13 | 20261263 | 0.55       | 0.000 | -0.305   | -0.734      | 0.077    |
| B   | 13 | 20261263 | 0.55       | 1.000 | 0.059    | 0.007       | 1.000    |
| C   | 13 | 20261263 | 0.55       | 0.158 | -0.007   | 0.011       | 0.077    |
| D   | 13 | 20261263 | 0.55       | 0.000 | -0.305   | -0.734      | 0.077    |
| A   | 13 | 20261364 | 0.55       | 0.000 | -1.158   | -0.590      | 0.077    |
| B   | 13 | 20261364 | 0.55       | 1.000 | 0.014    | 0.003       | 1.000    |
| C   | 13 | 20261364 | 0.55       | 0.118 | 0.018    | 0.016       | 0.077    |
| D   | 13 | 20261364 | 0.55       | 0.000 | -1.158   | -0.590      | 0.077    |
| A   | 13 | 20261465 | 0.55       | 0.000 | -0.333   | -1.076      | 0.077    |
| B   | 13 | 20261465 | 0.55       | 1.000 | 0.029    | 0.010       | 1.000    |
| C   | 13 | 20261465 | 0.55       | 0.197 | -0.009   | 0.016       | 0.077    |
| D   | 13 | 20261465 | 0.55       | 0.000 | -0.333   | -1.076      | 0.077    |

| Arm | n  | Seed     | Train frac | OOD   | Patch-CE | Wrong patch | Weakness |
|-----|----|----------|------------|-------|----------|-------------|----------|
| A   | 13 | 20261566 | 0.55       | 0.000 | -0.777   | -1.066      | 0.077    |
| B   | 13 | 20261566 | 0.55       | 1.000 | 0.021    | 0.008       | 1.000    |
| C   | 13 | 20261566 | 0.55       | 0.145 | 0.004    | 0.018       | 0.077    |
| D   | 13 | 20261566 | 0.55       | 0.000 | -0.777   | -1.066      | 0.077    |
| A   | 13 | 20261667 | 0.55       | 0.000 | -0.382   | -0.717      | 0.077    |
| B   | 13 | 20261667 | 0.55       | 1.000 | 0.016    | 0.002       | 1.000    |
| C   | 13 | 20261667 | 0.55       | 0.224 | 0.000    | 0.000       | 0.077    |
| D   | 13 | 20261667 | 0.55       | 0.000 | -0.382   | -0.717      | 0.077    |
| A   | 13 | 20261768 | 0.55       | 0.000 | -0.865   | -1.136      | 0.077    |
| B   | 13 | 20261768 | 0.55       | 1.000 | 0.013    | 0.005       | 1.000    |
| C   | 13 | 20261768 | 0.55       | 0.171 | -0.001   | 0.006       | 0.077    |
| D   | 13 | 20261768 | 0.55       | 0.000 | -0.865   | -1.136      | 0.077    |
| A   | 13 | 20261368 | 0.70       | 0.000 | -0.327   | -0.724      | 0.077    |
| B   | 13 | 20261368 | 0.70       | 1.000 | 0.058    | 0.003       | 1.000    |
| C   | 13 | 20261368 | 0.70       | 0.157 | 0.005    | 0.036       | 0.077    |
| D   | 13 | 20261368 | 0.70       | 0.000 | -0.327   | -0.724      | 0.077    |
| A   | 13 | 20261469 | 0.70       | 0.000 | -0.257   | -1.076      | 0.077    |
| B   | 13 | 20261469 | 0.70       | 1.000 | 0.020    | 0.003       | 1.000    |
| C   | 13 | 20261469 | 0.70       | 0.216 | 0.013    | 0.004       | 0.077    |
| D   | 13 | 20261469 | 0.70       | 0.000 | -0.257   | -1.076      | 0.077    |
| A   | 13 | 20261570 | 0.70       | 0.000 | -0.316   | -0.734      | 0.077    |
| B   | 13 | 20261570 | 0.70       | 1.000 | 0.006    | 0.004       | 1.000    |
| C   | 13 | 20261570 | 0.70       | 0.118 | 0.012    | 0.015       | 0.077    |
| D   | 13 | 20261570 | 0.70       | 0.000 | -0.316   | -0.734      | 0.077    |
| A   | 13 | 20261671 | 0.70       | 0.000 | -0.202   | -0.732      | 0.077    |
| B   | 13 | 20261671 | 0.70       | 1.000 | 0.031    | 0.001       | 1.000    |
| C   | 13 | 20261671 | 0.70       | 0.118 | 0.007    | 0.177       | 0.077    |
| D   | 13 | 20261671 | 0.70       | 0.000 | -0.202   | -0.732      | 0.077    |
| A   | 13 | 20261772 | 0.70       | 0.000 | -0.463   | -0.725      | 0.077    |
| B   | 13 | 20261772 | 0.70       | 1.000 | 0.031    | 0.005       | 1.000    |
| C   | 13 | 20261772 | 0.70       | 0.098 | 0.015    | 0.056       | 0.077    |
| D   | 13 | 20261772 | 0.70       | 0.000 | -0.463   | -0.725      | 0.077    |
| A   | 13 | 20261873 | 0.70       | 0.000 | -0.694   | -0.742      | 0.077    |
| B   | 13 | 20261873 | 0.70       | 1.000 | 0.054    | 0.004       | 1.000    |
| C   | 13 | 20261873 | 0.70       | 0.118 | -0.019   | 0.089       | 0.077    |
| D   | 13 | 20261873 | 0.70       | 0.000 | -0.694   | -0.742      | 0.077    |

**A.2.3 E4.** All 108 Pythia LoRA cells from experiments/commitment\_surface/results/e4\_pythia\_lora\_v2\_appendix.json. No requested appendix fields are unavailable.

| Size | Arm | n  | Seed     | OOD   | Ablated OOD | Patch-CE | Weakness | Wrong group |
|------|-----|----|----------|-------|-------------|----------|----------|-------------|
| 70m  | A   | 13 | 20260709 | 0.000 | 0.000       | -2.678   | 0.077    | 0.077       |
| 70m  | B   | 13 | 20260709 | 0.714 | 0.000       | 4.703    | 0.077    | 0.077       |
| 70m  | C   | 13 | 20260709 | 0.143 | 0.000       | 4.140    | 0.077    | 0.077       |
| 70m  | D   | 13 | 20260709 | 0.000 | 0.000       | -2.678   | 0.077    | 0.077       |
| 70m  | A   | 13 | 20260809 | 0.000 | 0.143       | -4.267   | 0.077    | 0.077       |
| 70m  | B   | 13 | 20260809 | 0.857 | 0.143       | 4.995    | 0.077    | 0.077       |
| 70m  | C   | 13 | 20260809 | 0.000 | 0.143       | 1.327    | 0.077    | 0.077       |
| 70m  | D   | 13 | 20260809 | 0.000 | 0.143       | -4.267   | 0.077    | 0.077       |
| 70m  | A   | 13 | 20260909 | 0.000 | 0.000       | -0.952   | 0.077    | 0.077       |
| 70m  | B   | 13 | 20260909 | 0.857 | 0.000       | 5.097    | 0.077    | 0.077       |
| 70m  | C   | 13 | 20260909 | 0.000 | 0.000       | 2.638    | 0.077    | 0.077       |
| 70m  | D   | 13 | 20260909 | 0.000 | 0.000       | -0.952   | 0.077    | 0.077       |
| 70m  | A   | 17 | 20260709 | 0.000 | 0.000       | -2.509   | 0.059    | 0.059       |
| 70m  | B   | 17 | 20260709 | 0.667 | 0.000       | 4.223    | 0.059    | 0.059       |
| 70m  | C   | 17 | 20260709 | 0.000 | 0.000       | 1.092    | 0.059    | 0.059       |
| 70m  | D   | 17 | 20260709 | 0.000 | 0.000       | -2.509   | 0.059    | 0.059       |
| 70m  | A   | 17 | 20260809 | 0.000 | 0.000       | -3.455   | 0.059    | 0.059       |
| 70m  | B   | 17 | 20260809 | 0.889 | 0.000       | 5.711    | 0.059    | 0.059       |
| 70m  | C   | 17 | 20260809 | 0.222 | 0.000       | 3.404    | 0.059    | 0.059       |
| 70m  | D   | 17 | 20260809 | 0.000 | 0.000       | -3.455   | 0.059    | 0.059       |
| 70m  | A   | 17 | 20260909 | 0.000 | 0.111       | -2.797   | 0.059    | 0.059       |
| 70m  | B   | 17 | 20260909 | 0.889 | 0.111       | 5.092    | 0.059    | 0.059       |
| 70m  | C   | 17 | 20260909 | 0.111 | 0.111       | 1.837    | 0.059    | 0.059       |
| 70m  | D   | 17 | 20260909 | 0.000 | 0.111       | -2.797   | 0.059    | 0.059       |
| 70m  | A   | 23 | 20260709 | 0.000 | 0.091       | -1.068   | 0.043    | 0.043       |
| 70m  | B   | 23 | 20260709 | 0.909 | 0.091       | 5.986    | 0.043    | 0.043       |
| 70m  | C   | 23 | 20260709 | 0.000 | 0.091       | 0.628    | 0.043    | 0.043       |
| 70m  | D   | 23 | 20260709 | 0.000 | 0.091       | -1.068   | 0.043    | 0.043       |
| 70m  | A   | 23 | 20260809 | 0.000 | 0.000       | -0.342   | 0.043    | 0.043       |
| 70m  | B   | 23 | 20260809 | 1.000 | 0.000       | 6.474    | 1.000    | 0.043       |
| 70m  | C   | 23 | 20260809 | 0.091 | 0.000       | 1.870    | 0.043    | 0.043       |
| 70m  | D   | 23 | 20260809 | 0.000 | 0.000       | -0.342   | 0.043    | 0.043       |
| 70m  | A   | 23 | 20260909 | 0.000 | 0.000       | 0.646    | 0.043    | 0.043       |
| 70m  | B   | 23 | 20260909 | 0.818 | 0.000       | 6.017    | 0.043    | 0.043       |
| 70m  | C   | 23 | 20260909 | 0.091 | 0.000       | 2.270    | 0.043    | 0.043       |
| 70m  | D   | 23 | 20260909 | 0.000 | 0.000       | 0.646    | 0.043    | 0.043       |
| 160m | A   | 13 | 20260709 | 0.000 | 0.000       | -7.546   | 0.077    | 0.077       |
| 160m | B   | 13 | 20260709 | 0.714 | 0.000       | 3.415    | 0.077    | 0.077       |
| 160m | C   | 13 | 20260709 | 0.143 | 0.000       | 2.755    | 0.077    | 0.077       |
| 160m | D   | 13 | 20260709 | 0.000 | 0.000       | -7.546   | 0.077    | 0.077       |
| 160m | A   | 13 | 20260809 | 0.143 | 0.000       | -1.909   | 0.077    | 0.077       |
| 160m | B   | 13 | 20260809 | 1.000 | 0.000       | 5.396    | 1.000    | 0.077       |
| 160m | C   | 13 | 20260809 | 0.000 | 0.000       | 1.645    | 0.077    | 0.077       |
| 160m | D   | 13 | 20260809 | 0.143 | 0.000       | -1.909   | 0.077    | 0.077       |
| 160m | A   | 13 | 20260909 | 0.000 | 0.000       | -1.527   | 0.077    | 0.077       |
| 160m | B   | 13 | 20260909 | 0.857 | 0.000       | 5.047    | 0.077    | 0.077       |
| 160m | C   | 13 | 20260909 | 0.143 | 0.000       | 2.816    | 0.077    | 0.077       |
| 160m | D   | 13 | 20260909 | 0.000 | 0.000       | -1.527   | 0.077    | 0.077       |
| 160m | A   | 17 | 20260709 | 0.556 | 0.000       | 4.414    | 0.059    | 0.059       |
| 160m | B   | 17 | 20260709 | 0.889 | 0.000       | 4.769    | 0.059    | 0.059       |
| 160m | C   | 17 | 20260709 | 0.000 | 0.000       | 1.646    | 0.059    | 0.059       |
| 160m | D   | 17 | 20260709 | 0.556 | 0.000       | 4.414    | 0.059    | 0.059       |
| 160m | A   | 17 | 20260809 | 0.000 | 0.000       | -2.674   | 0.059    | 0.059       |
| 160m | B   | 17 | 20260809 | 0.889 | 0.000       | 5.465    | 0.059    | 0.059       |
| 160m | C   | 17 | 20260809 | 0.111 | 0.000       | 1.369    | 0.059    | 0.059       |
| 160m | D   | 17 | 20260809 | 0.000 | 0.000       | -2.674   | 0.059    | 0.059       |
| 160m | A   | 17 | 20260909 | 0.000 | 0.111       | -4.009   | 0.059    | 0.059       |

| Size | Arm | n  | Seed     | OOD   | Ablated OOD | Patch-CE | Weakness | Wrong group |
|------|-----|----|----------|-------|-------------|----------|----------|-------------|
| 160m | B   | 17 | 20260909 | 0.889 | 0.111       | 4.308    | 0.059    | 0.059       |
| 160m | C   | 17 | 20260909 | 0.111 | 0.111       | 1.140    | 0.059    | 0.059       |
| 160m | D   | 17 | 20260909 | 0.000 | 0.111       | -4.009   | 0.059    | 0.059       |
| 160m | A   | 23 | 20260709 | 0.000 | 0.000       | -1.967   | 0.043    | 0.043       |
| 160m | B   | 23 | 20260709 | 1.000 | 0.000       | 5.735    | 1.000    | 0.043       |
| 160m | C   | 23 | 20260709 | 0.000 | 0.000       | 1.956    | 0.043    | 0.043       |
| 160m | D   | 23 | 20260709 | 0.000 | 0.000       | -1.967   | 0.043    | 0.043       |
| 160m | A   | 23 | 20260809 | 0.000 | 0.000       | 0.199    | 0.043    | 0.043       |
| 160m | B   | 23 | 20260809 | 1.000 | 0.000       | 5.849    | 1.000    | 0.043       |
| 160m | C   | 23 | 20260809 | 0.091 | 0.000       | 1.057    | 0.043    | 0.043       |
| 160m | D   | 23 | 20260809 | 0.000 | 0.000       | 0.199    | 0.043    | 0.043       |
| 160m | A   | 23 | 20260909 | 0.000 | 0.000       | -1.036   | 0.043    | 0.043       |
| 160m | B   | 23 | 20260909 | 0.818 | 0.000       | 5.527    | 0.043    | 0.043       |
| 160m | C   | 23 | 20260909 | 0.091 | 0.000       | 1.714    | 0.043    | 0.043       |
| 160m | D   | 23 | 20260909 | 0.000 | 0.000       | -1.036   | 0.043    | 0.043       |
| 410m | A   | 13 | 20260709 | 0.000 | 0.000       | -0.434   | 0.077    | 0.077       |
| 410m | B   | 13 | 20260709 | 0.714 | 0.000       | 3.171    | 0.077    | 0.077       |
| 410m | C   | 13 | 20260709 | 0.143 | 0.000       | 1.277    | 0.077    | 0.077       |
| 410m | D   | 13 | 20260709 | 0.000 | 0.000       | -0.434   | 0.077    | 0.077       |
| 410m | A   | 13 | 20260809 | 0.143 | 0.000       | 0.389    | 0.077    | 0.077       |
| 410m | B   | 13 | 20260809 | 1.000 | 0.000       | 4.085    | 1.000    | 0.077       |
| 410m | C   | 13 | 20260809 | 0.000 | 0.000       | -0.108   | 0.077    | 0.077       |
| 410m | D   | 13 | 20260809 | 0.143 | 0.000       | 0.389    | 0.077    | 0.077       |
| 410m | A   | 13 | 20260909 | 0.143 | 0.000       | 2.405    | 0.077    | 0.077       |
| 410m | B   | 13 | 20260909 | 0.857 | 0.000       | 3.931    | 0.077    | 0.077       |
| 410m | C   | 13 | 20260909 | 0.143 | 0.000       | 0.533    | 0.077    | 0.077       |
| 410m | D   | 13 | 20260909 | 0.143 | 0.000       | 2.405    | 0.077    | 0.077       |
| 410m | A   | 17 | 20260709 | 1.000 | 0.222       | 4.142    | 1.000    | 0.059       |
| 410m | B   | 17 | 20260709 | 1.000 | 0.222       | 4.148    | 1.000    | 0.059       |
| 410m | C   | 17 | 20260709 | 0.000 | 0.222       | 0.598    | 0.059    | 0.059       |
| 410m | D   | 17 | 20260709 | 1.000 | 0.222       | 4.142    | 1.000    | 0.059       |
| 410m | A   | 17 | 20260809 | 0.000 | 0.000       | 1.016    | 0.059    | 0.059       |
| 410m | B   | 17 | 20260809 | 0.889 | 0.000       | 4.384    | 0.059    | 0.059       |
| 410m | C   | 17 | 20260809 | 0.111 | 0.000       | 1.102    | 0.059    | 0.059       |
| 410m | D   | 17 | 20260809 | 0.000 | 0.000       | 1.016    | 0.059    | 0.059       |
| 410m | A   | 17 | 20260909 | 0.333 | 0.000       | 1.286    | 0.059    | 0.059       |
| 410m | B   | 17 | 20260909 | 0.889 | 0.000       | 4.136    | 0.059    | 0.059       |
| 410m | C   | 17 | 20260909 | 0.000 | 0.000       | 0.110    | 0.059    | 0.059       |
| 410m | D   | 17 | 20260909 | 0.333 | 0.000       | 1.286    | 0.059    | 0.059       |
| 410m | A   | 23 | 20260709 | 0.000 | 0.000       | -0.022   | 0.043    | 0.043       |
| 410m | B   | 23 | 20260709 | 0.909 | 0.000       | 4.368    | 0.043    | 0.043       |
| 410m | C   | 23 | 20260709 | 0.000 | 0.000       | -0.190   | 0.043    | 0.043       |
| 410m | D   | 23 | 20260709 | 0.000 | 0.000       | -0.022   | 0.043    | 0.043       |
| 410m | A   | 23 | 20260809 | 0.455 | 0.000       | 2.417    | 0.043    | 0.043       |
| 410m | B   | 23 | 20260809 | 1.000 | 0.000       | 4.661    | 1.000    | 0.043       |
| 410m | C   | 23 | 20260809 | 0.091 | 0.000       | 0.441    | 0.043    | 0.043       |
| 410m | D   | 23 | 20260809 | 0.455 | 0.000       | 2.417    | 0.043    | 0.043       |
| 410m | A   | 23 | 20260909 | 0.273 | 0.000       | 2.241    | 0.043    | 0.043       |
| 410m | B   | 23 | 20260909 | 0.909 | 0.000       | 4.576    | 0.043    | 0.043       |
| 410m | C   | 23 | 20260909 | 0.091 | 0.000       | -0.056   | 0.043    | 0.043       |
| 410m | D   | 23 | 20260909 | 0.273 | 0.000       | 2.241    | 0.043    | 0.043       |

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### A.4 Reviewer response (pre-empt)

We anticipate three lines of criticism and answer here.

**R1. "Your cyclic modular addition is a synthetic world; the E4 external contact is still on modular addition, so this isn't external."**

Response. The E4 external is *not* another synthetic world. It uses a public open-weights model family (Pythia 70m/160m/410m) trained on The Pile, and asks whether a LoRA fine-tune with cyclic-orbit augmentation produces load-bearing OOD generalization. The old-frame reading of our own prior program predicted a yes on weakness readout; our P1 hard-kill and E4 Arm A both said no. The commitment-first reframe predicted that Arm B (train-time compat intervention) would recover, and E4 does



— at  $n=13$  in smoke, at scale in the full grid. This is exactly the kind of *pre-registered directional prediction on a system the lab did not build* the prior program's critique identified as missing. It is not the last word on external contact — we say so in Section 6.4 — but it is a genuine one.

## **R2. "Patch-CE is confounded with total LoRA effect; you're just measuring that the model uses fine-tuning."**

Response. The wrong-group Arm C is the anti-cheat. Arm C has the same *augmentation volume* as Arm B — same number of extra training pairs, same LoRA capacity, same optimizer trajectory — but the augmented pair  $(x, y = \text{truth}(x))$  is transformed by a random non-cyclic permutation  $\pi$  to  $(\pi(x), \pi(y))$ , teaching the model the wrong equivariance  $f(\pi(x)) = \pi(f(x))$  instead of the cyclic action  $f(x + k) = f(x) + k$ . On any input that also appears in the base training set with its true label, Arm C's augmentation is *inconsistent* with the cyclic rule, so the model cannot fit both without choosing. If patch-CE were only measuring "the LoRA update is used", Arms B and C would have similar patch-CE and similar OOD. Empirically the anti-cheat holds: LoRA is only load-bearing when the augmentation *group* matches the deployment generator, not merely when augmentation *volume* is present. However, Arm C controls volume, **not label exposure**: Arm B's cyclic augmentation places correctly labeled examples on the held-out deployment support, while Arm C places incorrectly labeled ones there. An earlier revision of Arm C used the labeled coverage augmentation  $(\pi(x), b) \rightarrow \text{truth}(\pi(x), b)$ , which just extends training coverage with correct labels and — as expected — also produced OOD generalization. That result answers a different question ("does adding correct-labeled coverage help?" — yes) and, importantly, keeps the coverage explanation of Arm B live: the reviewer's confound is real, not hypothetical. Section 6.5 states the confound explicitly and pre-registers the severe follow-up (generator regularization confined to train support, tested on a novel group element or modulus) with kill criteria. Until it runs, R2's strongest defensible reading of E4 is "aligned train-time intervention recovers external OOD where readout selection does not," with the mechanism not yet isolated.

## **R3. "The anti-Goodhart control loop reads like a philosophical compression, not an empirical result."**

Response. It is a compression *hypothesis* over the prior Correction Chain, made testable by the load-bearing subset claim: {allocate, cool, reopen} carry the causal load; {detect, saturate} are diagnostic. The claim is falsifiable by a factorial ablation in Suite C — drop any one of {allocate, cool, reopen} and check whether the world-change re-engagement gate survives. We do not run that factorial in this paper (F4 in Section 4 is honest about this), but it is the concrete follow-up and we name it. The paper's headline claims (C1–C3 and C5) do not depend on M4; M4 is a load-bearing conjecture whose downstream test is designed but not yet completed.